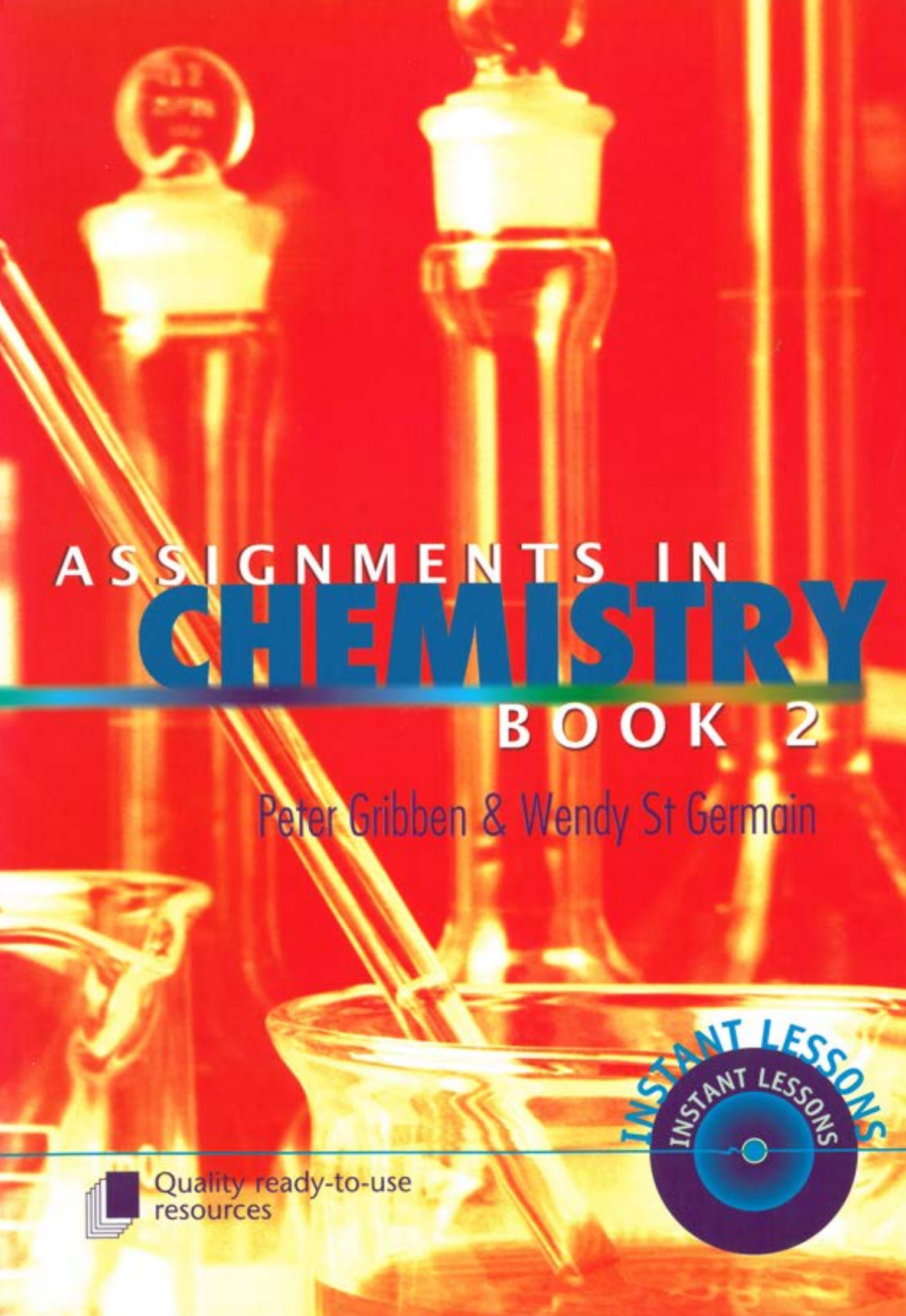




ASSIGNMENTS IN **CHEMISTRY** BOOK 2

Peter Gribben & Wendy St Germain



Quality ready-to-use
resources



The background of the cover is a black and white photograph of chemistry glassware. In the foreground, a glass beaker is partially filled with liquid, and a glass rod is stirring it. Behind it, several other glass vessels, including a volumetric flask and a graduated cylinder, are visible, some containing liquids. The lighting creates highlights and shadows on the glass surfaces.

ASSIGNMENTS IN
CHEMISTRY
BOOK 2

Peter Gribben & Wendy St Germain



Wendy St Germain undertook and completed her science training at Sydney's Macquarie University, majoring in zoology and genetics and graduating with distinctions in several courses. Her ability to write science clearly and in ways which clarify rather than confuse students has made her a popular author. Wendy has written many titles, both independently and as a co-author with Peter Gribben. The subjects of her books include genetics, biology, physics, chemistry, space science, aboriginal studies, english and more. She has written numerous other educational materials including educational CD ROMs, integrated units, 'in class' tests and sections of the NSW Independent Trial Examinations. Wendy has recently developed a whole new following with her children's fiction and young adult titles.

Peter Gribben is currently Head Teacher of Science and Agriculture at Crookwell High School in the Southern Tablelands of New South Wales. He has taught Science at all levels for nearly 30 years in England, the Middle East and Australia. Peter has written many science books; both conventional text-books and photocopiable worksheets. He also has considerable experience in preparing examination material. Peter has taught senior chemistry in both Victoria and New South Wales. He has a special interest in preparing chemistry (and science) materials that are accessible to students of all levels.

Peter's personal interests include raising German Shepherds, restoring Minis and reading science fiction.

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Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

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An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
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INTRODUCTION

This series of chemistry assignments has been written by a highly-experienced team of authors. Each worksheet has been specially designed to meet the needs of busy teachers of chemistry – they require minimal preparation and do not compromise on quality of content. Most of the material requires very little prior knowledge from the student. New chemical terms are explained as they are met in the text and important concepts are revised or summarised. Areas that students have traditionally found difficult have been broken down into sections and are examined one step at a time.

There are many worked examples of calculations and sufficient questions for students to practise and master difficult concepts. Research questions have been included where appropriate to give students the opportunity to explore the history of chemistry.

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Many everyday names represent a fixed number of objects. For example:

- a soccer team: 11
- a netball team: 7
- a hockey team: 11
- an Australian Rules football team: 18
- a dozen eggs: 12
- a gross of wood screws: 144
- a ream of paper: 500 sheets
- a century of runs: 100

- 1 Give another name that stands for a fixed quantity.

Chemistry is all about objects like atoms, molecules and ions. These objects are all so small that there must be many billions of them before they can be seen. A special number used in chemistry is 6.02×10^{23} , called the **mole**. It is also called **Avogadro's number**, in honour of Amedeo Avogadro (1776–1856). The mole is simply a convenient unit of measurement that chemists use because it fits in with 'natural' numbers. It is like saying there are 365 days in a year, the diameter of the Earth is 12 800 km, light travels at 300 000 kilometres per second (approximately), and so on.

Units of measurement are often used in mathematics questions:

EXAMPLES

- i How many dozens (units of 12) are there in 36 units?

Divide the total number by the number in one dozen: $36 \div 12 = 3$ dozen.

- ii How many days are there in 100 hours?

$100 \div 24 = 4.17$ days

The same mathematic principles apply to calculations with moles.

EXAMPLES

- iii How many atoms of sodium are there in 0.5 moles of sodium?

'Of' is a hint to use multiplication. Multiply the number of moles by the number of atoms in a mole:
 $0.5 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$

- iv How many moles of water molecules are in 57×10^{27} particles of water?

Divide the total number of particles by the number of particles that makes up a mole:
 $57 \times 10^{27} \div 6.02 \times 10^{23} = 9.5 \times 10^4$

(Note that when using moles we sometimes get 'silly' or 'ridiculously large' numbers.)

- 2 Calculate how many moles are present in each of the following.

- a 88×10^{23} copper atoms b 12×10^{23} helium atoms

Masses of chemicals

Atomic mass units (amu) are used to describe the mass of atoms. Today, atomic mass units are defined in terms of the ^{12}C isotope, which is assigned a mass of exactly 12.00 amu. The amu is too small a unit to use for everyday objects: 1 amu is equivalent to 1.67×10^{-27} grams.

- 3 Explain what is meant by 'the isotope ^{12}C '.

Chemists need to know what mass of an element will give one mole of atoms of that element. The mole can be defined as the number of atoms in 12.00 grams of the ^{12}C isotope. That is, in 12.00 grams of the ^{12}C isotope there are 6.0221367×10^{23} atoms (commonly abbreviated to 6.02×10^{23}).

In a Periodic Table, carbon is assigned a slightly different mass: 12.01. This is because naturally occurring carbon is a mixture of isotopes; 12.01 is an average of the masses of the isotopes according to the proportion in which they occur in nature. The Periodic Table quotes hydrogen as having a mass of 1.008. This means that atoms of naturally occurring hydrogen have an average mass of 1.008 amu; therefore a mole of hydrogen atoms has a mass of 1.008 g. Helium is quoted as having a mass of 4.003. This means that atoms of naturally occurring helium have an average mass of 4.003 amu; therefore a mole of helium atoms has a mass of 4.003 g.

A mole of atoms of any element = the atomic mass of that element in grams.

4 Give the chemical symbol, atomic mass and the mass of a mole of the following atoms:

a oxygen

b neon

c sodium

d copper

Formula mass

The atomic mass of a molecule or compound is not referred to in chemistry because more than one atom is involved and each atom making up the molecule has its own atomic mass. Instead, the **formula mass** is used because each molecule has a specific formula. The formula mass is calculated by adding the atomic masses of each element in the molecule.

EXAMPLE

v What is the mass of one mole of bromine?

Bromine exists as a diatomic molecule, Br_2 , where two bromine atoms make up one bromine molecule. Each bromine atom has a mass of 80.00 amu. Therefore, the formula mass of a bromine molecule is 160.00 (2×80.00) and one mole of bromine molecules has a mass of 160.00 g.

5 Calculate the masses of one mole of the following diatomic molecules:

a oxygen (O_2) [$\text{O} = 16.00$ amu]

b nitrogen (N_2) [$\text{N} = 14.01$ amu]

c chlorine (Cl_2) [$\text{Cl} = 35.45$ amu]

The same principle can be applied to work out the masses of molecules.

EXAMPLES

vi What is the mass of a mole of magnesium chloride?

In magnesium chloride (MgCl_2) there is one magnesium atom (24.31) and two chlorine atoms ($35.45 \times 2 = 70.90$) present in the formula.

Hence the formula mass of magnesium chloride is $24.31 + 70.90 = 95.21$. This is the total of all of the atomic masses of the atoms that are shown in the formula of the substance. Therefore the mass of a mole of magnesium chloride is 95.21 g.

vii What is the mass of a mole of water?

Water has the formula H_2O . Each molecule contains two hydrogen atoms and one oxygen atom. Hydrogen atoms have a mass of 1.008 amu and oxygen atoms a mass of 16.00 amu.

The formula mass is $(2 \times 1.008) + (1 \times 16.00) = 18.02$ amu. Therefore a mole of water molecules has a mass of 18.02 g.

6 Calculate the mass of one mole of each compound. Refer to a Periodic Table for the relevant atomic masses.

a carbon dioxide, CO_2

b phosphorus pentachloride, PCl_5

c aluminium oxide, Al_2O_3

d glucose, $\text{C}_6\text{H}_{12}\text{O}_6$

e sulfuric acid, H_2SO_4

f calcium hydroxide, $\text{Ca}(\text{OH})_2$

g ethanoic acid, CH_3COOH

Often parts or multiples of moles are used in chemistry. To calculate these, work out the molar mass of the substance and then multiply it by the number of moles.

EXAMPLE

viii What is the mass of 0.75 moles of water?

The mass is the number of moles multiplied by molar mass: $0.75 \times 18.02 = 13.52$ g

7 For each of the following, give the formula and calculate the mass in grams.

- a 0.5 moles of carbon dioxide
- b 0.1 moles of hydrogen chloride
- c 2 moles of sulfuric acid
- d 0.1 moles of phosphorus pentachloride
- e 0.25 moles of magnesium nitrate
- f 1.5 moles of ammonia
- g 1 mole of chlorine molecules

Chemists need to work out how many moles of a substance are present in a given mass. To do this, calculate the mass of one mole (atomic, molecular or formula mass in grams). Then divide the given mass by the mass of one mole.

$$\text{number of moles} = \frac{\text{given mass}}{\text{molar mass}}$$

EXAMPLE

ix How many moles of ammonia molecules are present in 30.00 g of ammonia gas?

Ammonia has the formula NH_3 . Calculate its formula mass:

$$(1 \times \text{N}) + (3 \times \text{H}) = (1 \times 14.01) + (3 \times 1.008) = 17.03$$

One mole of ammonia has a mass of 17.03 g or 17.03 g of ammonia is one mole.

Divide the given mass by 17.03 to calculate the number of moles of ammonia: $30.00 \div 17.03 = 1.762$.

Given that there are 30.00 g of the substance and 17.03 g per mole, calculate the number of moles: $30.00 \div 17.03 = 1.762$. Therefore there are **1.762 moles** of ammonia in 30.00 g of ammonia gas.

8 Calculate how many moles are in each mass.

- a 12.0 g of helium atoms
- b 20.0 g of bromine atoms
- c 35.5 g of chlorine molecules
- d 20.0 g of sodium hydroxide
- e 200 g of calcium carbonate
- f 5.60 g of iron
- g 9.80 g of sulfuric acid

Equations and masses

Simple reaction equations can be deduced from relative masses of pure substances. First, find the number of moles present of each substance (divide mass present by formula mass) and then work out the ratio of moles of each substance.

EXAMPLE

x 7.95 g of copper (II) oxide reacts with 0.60 g of carbon to produce 6.35 g of copper and 2.2 g of carbon dioxide. Work out the equation for the reaction.

copper (II) oxide (CuO): formula mass = $63.55 + 16.00 = 79.55$ amu

carbon (C): atomic mass = 12.01 amu

carbon dioxide (CO_2): formula mass = $12.01 + (2 \times 16.00) = 12.01 + 32.00 = 44.01$ amu

No. of moles of copper oxide = $7.95 \div 79.55 = 0.1$ moles

No. of moles of carbon = $0.60 \div 12.01 = 0.05$ moles

No. of moles of carbon dioxide = $2.2 \div 44.01 = 0.05$ moles

The ratio $\text{CuO} : \text{C} : \text{CO}_2 = 0.1 : 0.05 : 0.05 = 2 : 1 : 1$

That is, the reaction equation is $2\text{CuO (s)} + \text{C (s)} \rightarrow 2\text{Cu (s)} + \text{CO}_2 \text{ (g)}$

9 Work out the equation for the reaction between 40.00 g of iron (III) oxide and 21.00 g of carbon monoxide, producing 28.00 g of iron and 33.00 g of carbon dioxide.

Empirical formulae

The empirical formula for a compound is the simplest ratio of atoms present. The molecular formula shows the actual number of atoms present. For some compounds the empirical and molecular formulae are the same. The empirical formula of hydrogen peroxide is HO; there is one hydrogen atom for every oxygen atom. The molecular formula of hydrogen peroxide is actually H_2O_2 ; there are two atoms of hydrogen and two atoms of oxygen in every molecule of hydrogen peroxide.

The empirical formula of a compound is the simplest formula that shows the ratio of the different atoms in the compound.

10 Write the empirical formula for each substance.

- a glucose, $\text{C}_6\text{H}_{12}\text{O}_6$
- b butane, C_4H_{10}
- c ethanoic acid, CH_3COOH
- d aluminium oxide, Al_2O_3

The empirical formula and the molecular mass can be used to find the molecular formula of some substances.

EXAMPLE

- xi** A hydrocarbon has an empirical formula of CH_2 . It has a formula mass of 84.00. What is its molecular formula?

There are two hydrogens for every carbon atom, which means the molecular formula is a multiple of CH_2 , such as C_2H_4 or C_3H_6 . This can be written as $(\text{CH}_2)_n$.

CH_2 is one carbon atom (1×12.01) plus two hydrogen atoms (2×1.008) = $12.01 + 2.016 = 14.02$.

To find the number of CH_2 units, divide the formula mass by the mass of the empirical formula:
 $84.00 \div 14.02 = 6$. This means that there are 6 groups of CH_2 in a molecule of the hydrocarbon, so the molecular formula is C_6H_{12} .

- 11** A hydrocarbon has twice as many hydrogen atoms as carbon atoms. Its relative molecular mass is approximately 112 amu.

- a Calculate its empirical formula.
- b Calculate its molecular formula.

The relative masses of the elements in a compound depend on the masses and the relative numbers of the atoms present. If the total mass of a compound and the individual masses of the elements that make it up are known, the relative numbers of atoms of each element present can be worked out. Divide the masses of elements present by their relative atomic masses (ram). Then find the ratio of these numbers. In other words, use the number of moles of different atoms present in a substance to work out the empirical formula of that substance.

EXAMPLE

- xii** 11.00 g of a compound of mercury and chlorine contained 9.34 g of mercury. The relative molecular mass (rmm) was found to be 472.00 amu.

- Calculate the empirical formula of the compound.
- Calculate the molecular formula of the compound.
- Calculate the mass of the reactants: mercury = 9.34 g; chlorine = $(11.00 - 9.34) = 1.66$ g
- Calculate the number of moles:
 Mercury (Hg) has a ram of 200.6 amu. No. of moles = $9.34 \div 200.6 = 0.046$ moles.
 Chlorine (Cl) has a ram of 35.50 amu. No. of moles = $1.66 \div 35.50 = 0.046$ moles
- Calculate the ratio. 0.046 : 0.046 or 1 : 1
 Hence the empirical formula is HgCl.
- Use the empirical formula to calculate rmm: $200.60 + 35.45 = 236.05$ amu
- Work out the molecular formula:
 Actual rmm = 472.00 amu. $472.00 \div 236.10 = 2$. Hence the molecular formula is Hg_2Cl_2 .

- xiii 1.32 g of magnesium was dissolved in dilute hydrochloric acid. The solution was then heated in a stream of hydrogen chloride. 5.26 g of the anhydrous metal chloride was formed. Find the simplest formula for the metal chloride.

1.32 g of magnesium combined with $(5.26 - 1.32)$ g of chlorine = 3.94 g of chlorine

Moles of Mg atoms : moles of Cl atoms

$$= (1.32 \div 24) : (3.94 \div 35.5)$$

$$= 0.055 : 0.111 = 1 : 2 \text{ (to the nearest whole number)}$$

Therefore the simplest formula is MgCl_2 .

- 12 Two oxides of nitrogen were analysed. Both oxides contain two atoms of nitrogen per molecule. The first oxide has a rmm of 76 amu and the other oxide has a rmm of 108 amu. Calculate the molecular formula of each compound. Show your working.

(Hint: work out how much nitrogen is present per molecule, the remainder must be oxygen.)

- 13 0.48 g of magnesium is burned in oxygen to form 0.80 g of magnesium oxide.

- How much oxygen is used in the combustion?
- What is the empirical formula of magnesium oxide? Show your working.

Percentages and empirical formula

If the percentages (by mass) of elements in a substance are known, the empirical formula of the substance can be calculated.

EXAMPLE

- xvi A chemical analysis of an organic compound showed that it was made up of carbon (40%), hydrogen (6.7%) and oxygen (53.3%). What is its empirical formula?

Elements present:	C	H	O
Percentage:	40	6.6	53.3
ram:	12.01	1.008	16.00
percentage $\div$ ram	$40 \div 12.01$ $= 3.33$	$6.6 \div 1.008$ $= 6.66$	$53.3 \div 16.00$ $= 3.33$
divide by lowest common denominator (3.33)	1	2	1

Hence the empirical formula is CH_2O .

- 14 A compound has an empirical formula CH_2O . Its relative molecular mass (rmm) is 180.20 amu. Work out its molecular formula.

- 15 Car radiators often contain antifreeze. One particular type of antifreeze was analysed and found to contain carbon (38.7%), hydrogen (9.7%) and oxygen (51.6%).

- Calculate its empirical formula.
- Given that its rmm is 62.10 amu, what is its molecular formula?

- 16 Hydrated copper sulfate crystals were analysed and gave this percentage composition by mass: Cu = 25.8%, S = 12.8%, O = 25.6% and water of crystallisation = 36.1%. What is the empirical formula of the crystals? (Hint: treat the water of crystallisation by itself.)

- 17 Mark each statement as true or false.

- A mole of particles consists of Avogadro's number.
- The empirical formula of a substance shows the actual numbers and types of atom present.
- Relative molecular mass (rmm) is measured in atomic mass units (amu).
- The term 'mole' comes from the abbreviation for Molekulargewicht (molecular weight).
- Avogadro's number was discovered by Amedeo Avogadro.

One mole of any (ideal) gas occupies the same volume as any other (ideal) gas that is at the same temperature and pressure. **Ideal gases** are those that behave in a certain, predictable manner. Often a standard temperature and pressure (STP) or standard laboratory conditions (SLC) are quoted in reference to gases. Common standards include 1 atmosphere pressure (sometimes quoted as 101.3 kPa) and either 0°C (273 K) or 25°C (298 K).

For questions on this page assume that 1 mole of a gas at STP occupies 22.40 litres. Different sources will use different values so check carefully the values you are given to work with.

Quantities of gases can be quoted using moles, volume or mass. It is essential to be able to convert between these measurements.

$$\begin{aligned}\text{number of moles} &= \frac{\text{mass present}}{\text{molar mass}} \\ &= \frac{\text{volume present}}{\text{molar volume}}\end{aligned}$$

EXAMPLES

i How many moles of gas are present in 12.00 g of carbon dioxide?

Carbon dioxide has the formula CO_2 : one carbon atom (12×12.01) and two oxygen atoms (2×16.00)

Formula mass: $(12.01 + 32.00) = 44.01$ amu

Hence one mole of carbon dioxide has a mass of 44.01 g.

The number of moles is calculated by dividing the mass present (12.00 g) by the molar mass (44.01 g) = 0.2727 moles

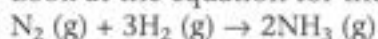
ii How many moles of sulfur dioxide are in 500 mL of the gas at STP?

Assume that one mole of sulfur dioxide occupies 22.40 litres. The number of moles is calculated by dividing the volume present (500 mL = 0.5 L) by the molar volume (22.40 L) = 0.022 moles.

- 1 Calculate the moles of gas present in 15.50 g of ammonia (NH_3).
- 2 Calculate the mass of 0.25 moles of oxygen molecules.
- 3 Calculate the number of moles (at STP) of 1.76 litres of sulfur trioxide.
- 4 Calculate the mass of 12.00 litres of hydrogen molecules. (Hint: you need to do two calculations.)
- 5 What would be the volume of 3.00 kg of butane (C_4H_{10}) at STP?

At constant temperature and pressure, the volume of a gas is proportional to the number of moles; therefore chemical equations involving gases deal with volumes of gas as well as numbers of moles.

Look at the equation for the manufacture of ammonia from nitrogen and hydrogen:



This equation can be read in several ways:

- One molecule of nitrogen reacts with three molecules of hydrogen to form two molecules of ammonia
- One mole of nitrogen reacts with three moles of hydrogen to form two moles of ammonia
- One volume of nitrogen reacts with three volumes of hydrogen to form two volumes of ammonia
- One litre of nitrogen will react with three litres of hydrogen to form two litres of ammonia.

6 Methane, CH_4 , ('natural gas') is used as a fuel in household gas heaters and stoves. Methane burns in air to form carbon dioxide and water.

- a Write a balanced equation to show the combustion of methane.
- b How many moles of oxygen are needed to burn one mole of methane?
- c How many litres of carbon dioxide are formed when 100 litres of methane is burned?

7 Sulfur dioxide, SO_2 , can react with oxygen to form sulfur trioxide, SO_3 .

- a Write a balanced equation for the reaction between sulfur dioxide and oxygen.
- b How many moles of oxygen are needed per mole of sulfur dioxide?
- c How many litres of oxygen are needed to react with 50 litres of sulfur dioxide?

Many chemical reactions occur in solutions. There are two parts to a solution: a **solute** and a **solvent**. The solute is the substance being dissolved. The solvent is the substance in which the solute is being dissolved. Water is a very common solvent. Concentrations of solutions can be measured in two ways:

- mass/mass
- mass/volume

Mass/mass states how much of a solute is dissolved in a given mass of solution. A solution of magnesium chloride (MgCl_2) can be described as 5 g/100 g. This means that 5 g of the solute (magnesium chloride) has been dissolved in 100 g of solution.

Mass/volume states how much of a solute is dissolved in a given volume of a solution. To make comparing solutions easier, the volume used is usually one litre. A solution of magnesium chloride stated to be 50 g L^{-1} has 50 grams of solute dissolved in each litre of solution.

- Express each concentration in grams per litre (g L^{-1}).
 - 50 g in 100 mL
 - 5 g in 500 mL
 - 25 g in 250 mL
 - 75 g in 2000 mL

Sometimes parts per million (ppm) are used to describe concentrations of solutions. For example, a salt water solution quoted as being 5 ppm salt would be $5 \div 1\,000\,000$ or 0.0005% salt.

- The active ingredient in a medication is quoted as 120 ppm. How many grams of this substance are in a 50 g tube of the medication? Show your working.

The mole concept is often used in solutions in terms of the **molarity** of a solution. Molarity means the number of moles of solute dissolved per litre of a solution and is sometimes abbreviated to M. A solution containing 1 mole of solute per litre of solution is called a 1.0 molar solution or 1.0 M. It can also be shown as 1.0 mol L^{-1} .

A mole of any compound is its **formula mass** expressed in grams.

The molarity of a solution can be calculated using the equation: $C = n \div V$ where C = concentration (moles per litre), n = number of moles and V = volume (number of litres).

EXAMPLE

- What is the molarity of a solution 42.40 g of sodium carbonate in 500 mL of water?

Sodium carbonate = Na_2CO_3

Formula mass = $(2 \times 22.99) + (1 \times 12.01) + (3 \times 16.00) = 45.98 + 12.01 + 48.00 = 106.00 \text{ amu}$

Hence 1 mole = 106.00 g

Actual mass = 42.40 g. Hence number of moles = $42.40 \div 106.00 = 0.40$ moles

Express in moles per litre: 0.40 moles is dissolved in 0.50 litre (500 mL); hence the solution is: $0.40 \div 0.50 = 0.80$ molar

- Calculate the mass of sodium chloride (NaCl) dissolved in 1 litre of a 5 molar solution.
- Calculate the mass of sodium hydroxide (NaOH) dissolved in 300 mL of a 2 molar solution.
- Calculate the molarity of each solution.
 - 9.52 g of magnesium chloride (MgCl_2) dissolved in 1 litre of solution.
 - 9.8 g of sulfuric acid (H_2SO_4) dissolved in 500 mL of solution.
 - 21.2 g of sodium carbonate (Na_2CO_3) dissolved in 2 litres of solution.
- Calculate the mass of sulfuric acid in 250 mL for each given concentration.
 - 1.0 molar solution
 - 0.5 molar solution
 - 3.0 molar solution
- Explain the concentration descriptions *mass/mass* and *mass/volume*.
- Define *molarity*.

A general (but not infallible) rule says **like dissolves like**. This means that polar solvents tend to dissolve polar/ionic substances and non-polar solvents tend to dissolve non-polar substances.

Remember, in a solution:

- the solvent and solute particles are evenly distributed (homogenous) throughout the solution
- the solute particles are too small to be seen
- there are forces of attraction between solvent and solute particles.

Before a substance can dissolve in water the forces between its particles and between water molecules must be broken. This results in new bonds being formed between the water and the solute. Breaking bonds requires energy to be put in (endothermic; *endo*: within); making bonds gives energy out (exothermic; *exo*: outside). If a covalent molecule is very large (it can consist of tens of thousands of atoms, e.g. cellulose and polythene) or is non-polar, it will not dissolve in water. The bonding in covalent network structures (e.g. diamond and silicon dioxide) is too strong to be overcome by any possible bonds between particles in the structure and water molecules.

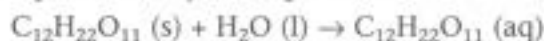
- 1 Suggest why a lattice network is so much stronger than other forms of bonding.
- 2 'A substance will only dissolve in a solvent if the attractive forces between solute and solvent particles are similar to those between the solvent/solvent and solute/solute particles.' Explain this statement in your own words.
- 3 Sodium chloride, NaCl, is soluble in water. Its structure is shown in figure 1.
 - a Is sodium chloride a covalent compound or an ionic compound? How do you know?
 - b Is water a polar solvent or a non-polar solvent? Explain your answer.

When positive and negative ions from an ionic lattice are separated, this process is called **dissociation** (*dissociate*, to part). The dissociated ions are surrounded by water molecules; they are **hydrated**. This can be shown by the equation:



- 4 Write an equation to show how each ionic solid dissolves and dissociates in water.
 - a potassium fluoride (KF)
 - b magnesium chloride (MgCl_2)
 - c potassium carbonate (K_2CO_3)
- 5 Not all ionic compounds are soluble. Why might an ionic compound be insoluble?

Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$, sugar) is a soluble covalent molecule. Although it is covalent, it does have polar parts which can be attracted by water molecules, forming hydrogen bonds. The dissolving of sugar in water can be represented by the equation:



- 6 Both sugar (sucrose) and common salt (sodium chloride) dissolve. How does the dissolving of sugar differ from that of sodium chloride?

Some covalent compounds are very highly polarised. Figure 2 shows the structure of hydrogen chloride. The hydrogen is relatively positive and the chlorine is relatively negative. The covalent bonds in polar molecules can be broken by the electrostatic forces between water and solute molecules. New bonds are formed. One example is when hydrogen chloride dissolves and ionises in water:



(Note that H_3O^+ is known as the hydronium ion; it is responsible for acidic properties.)

- 7 What is an aqueous solution of hydrogen chloride called?
- 8 A similar reaction to that of hydrogen chloride dissolving in water occurs when liquid nitric acid (HNO_3) dissolves in water. Represent this in a balanced equation.
- 9 Give two reasons why a covalent molecule might not dissolve in water.

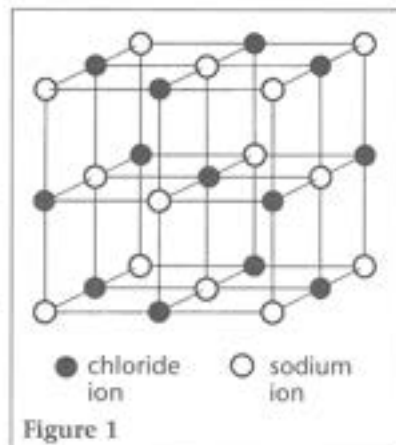


Figure 2

Solutions, suspensions, colloids and emulsions are all mixtures; they often involve particles scattered through a liquid.

A **solution** has particles (solute) mixed evenly throughout the medium (solvent). Although the dissolved particles can colour the solution, the individual solute particles cannot be seen. When a solution is passed through filter paper, the solute particles are small enough to fit through the tiny pores.

- 1 Write the missing word: 'When you mix a solid with a liquid and the solid disappears, the solid is said to have _____.'
- 2 Most of the solutions used in the laboratory are *aqueous*. Explain what is meant by *aqueous*.
- 3 Solutions are often described as *homogeneous*. Explain what is meant by *homogeneous*.
- 4 A student added copper sulfate to water and stirred it until no solid pieces could be seen. He said, 'The solid can't have dissolved because the mixture is coloured.' Discuss his statement.
- 5 What is meant by a *saturated solution*?

When a liquid can mix with water to form a solution, it is said to be **miscible** with water. (Liquids are only said to be miscible with other liquids, not solids or gases.) Ethanol (C_2H_5OH) and water are miscible.

- 6 What do you think two liquids that do not mix properly to form a solution are called?

A **suspension** has particles suspended in a medium. These particles do not all separate to mix evenly throughout the medium; they stay in clumps large enough to be seen with the naked eye. Suspensions tend to be cloudy until they settle out to form a sediment at the bottom of the container. The particles in suspensions do not pass through filter paper.

- 7 Give an example of a suspension.
- 8 Explain what is meant by 'a suspension settles out'.

A **colloid** is a type of suspension, but the particles do not settle out. The particles cannot be seen with the naked eye but can be seen through a microscope. The particles cannot pass through filter paper. Smoke in air and natural rubber are colloids. Sols are types of colloids; they are made up of very fine particles spread evenly throughout a fluid. An aerosol is a colloid where the liquid or solid particles are spread through a gas. The word *aerosol* is often used to describe the containers (spray cans or vaporisers) that produce an aerosol. Gels are colloids where the liquid medium is viscous and forms a jelly-like substance.

- 9 Name three substances that are packaged in aerosol containers.
- 10 Fog is similar to smoke; outline the similarities and differences between these two substances.
- 11 Name a type of gel and outline its use.
- 12 The term *medium* is often used in reference to mixtures. What is a medium?
- 13 Give a new example of a colloid.

The Tyndall effect

John Tyndall (1820–1893) was a British scientist who studied the scattering of light by dust and other small particles. This scattering of light is now called the **Tyndall effect**. If a beam of light is shone through a clear fluid in which small particles are suspended, the longer wavelengths of visible light (e.g. blue) are scattered less effectively than the shorter wavelengths of light (e.g. red). True solutions do not scatter light.

- 14 Explain whether the Tyndall effect would be shown in each substance.
 - a fog
 - b sugar dissolved in water

Research 15 Glass is a *super-cooled solution*. Find out what this means.

16 Robert Brown was one of the first people to study colloids and the first to describe a phenomenon now called Brownian motion. Outline what Brownian motion involves.

Many different items of glassware are used when handling liquids. Below are diagrams (not to scale), names and uses of glassware. Cut out each box and match each name and use to the correct diagram.

1		2		3		4	
5		6		7		8	
9		Dropper		Stirring rod		Conical flask	
		Test tube		Volumetric flask		Burette	
		Pipette		Funnel		Measuring cylinder	
		Beaker		A Stirs liquids.		B Delivers variable volumes very precisely by means of a tap	
10		C Delivers a fixed amount of liquid		D Can swirl contents without loss		E Helps to pour liquids into containers	
		F Delivers drops		G Makes standard solutions		H Used for small-scale reactions	
		I Can measure variable volumes accurately		J Used for transporting liquid; gives approximate volumes			

Table 1 shows the solubility of a range of compounds.

In addition, most ammonium (NH_4^+) compounds are soluble.

1 Name each compound and describe its solubility in water.

- CuSO_4
- CaCO_3
- AgCl
- NH_4OH
- $\text{Pb}(\text{NO}_3)_2$
- Li_3PO_4

Precipitates

Some ionic compounds are soluble in water. When they dissolve, they break up into aqueous ions, which are usually metal ions with a positive charge (cations) and non-metals or radicals with a negative charge (anions). In solution, cations and anions may react together to form insoluble substances called **precipitates** (*precipitate*: to separate and 'rain down'). When an ionic substance dissolves in water, the anions and cations move about in solution, constantly interacting. (Remember opposite electrical charges attract.) If two ions of opposite charge come together and react to form an insoluble substance then a precipitate is formed. When precipitates are formed the reaction is almost instantaneous. To see if a precipitate is formed, and if so, what it is called:

- work out what ions are in the solution
- work out possible combinations of ions
- identify the precipitate formed.

EXAMPLE

i When a silver nitrate (AgNO_3) solution is mixed with a sodium chloride (NaCl) solution a white precipitate forms. What is this precipitate? Write a net ionic equation for the reaction.

Work out the possible results of positive ions pairing with negative ions: the two possible compounds are silver chloride (AgCl) and sodium nitrate (NaNO_3). A table of solubilities (see table 1) tells that silver chloride (a silver halide) is insoluble and sodium nitrate is soluble. Hence the precipitate must be silver chloride.

Show all the ions present:



Rewrite, omitting the 'spectator ions': $\text{Ag}^+ (\text{aq}) + \text{Cl}^- (\text{aq}) \rightarrow \text{AgCl} (\text{s})$

2 For each pair of solutions, indicate if a precipitate will be formed and write an ionic equation for the relevant reaction. Include the name of any precipitate formed.

- Barium chloride (BaCl_2) and sodium nitrate (NaNO_3).
- Calcium chloride (CaCl_2) and lead (II) nitrate ($\text{Pb}(\text{NO}_3)_2$).
- Copper (II) sulfate (CuSO_4) and barium hydroxide ($\text{Ba}(\text{OH})_2$).

3 Outline how you could remove the precipitate from a mixture.

Calculate the theoretical yield of a precipitate, either in moles or grams: First identify the precipitate and write a net ionic equation. Then:

- balance the equation
- look at the ratio of reactants and products in the precipitation reaction
- look at the question to decide what quantities are needed
- carry out the calculations.

Anions	Soluble	Slightly soluble	Insoluble
hydroxides	Group I, Ba^{2+}	Ca^{2+} , Sr^{2+}	all others
carbonates	Group I except Li^+	Li^+	all others
hydrogen carbonates	most		all others
halides	most	Pb^{2+} (hot)	Pb^{2+} , Ag^+
nitrates	all		all others
sulfates	most	Ca^{2+} , Ag^+	Ba^{2+} , Pb^{2+}
sulfides	Groups I & II		all others
sulfites	Group I		all others
phosphates	Group I, except Li^+		all others

Table 1

EXAMPLE

ii What is the name, formula and number of moles of the precipitate formed between an excess of barium chloride solution and 100 mL of 0.5 mol L⁻¹ sodium sulfate solution?

- BaCl₂ gives Ba²⁺ and 2 × Cl⁻. Na₂SO₄ gives 2 × Na⁺ and 1 × SO₄²⁻.
- Combinations include sodium chloride and barium sulfate.
- NaCl is soluble and BaSO₄ is insoluble, hence BaSO₄ is the precipitate.
- Ba²⁺ (aq) + SO₄²⁻ (aq) → BaSO₄ (s) (Spectator ions omitted)
- Ratio is 1:1, hence the number of moles of barium sulfate formed will equal the number of moles of sodium sulfate added.

Find the amount of barium sulfate formed:

- Number of moles = concentration (moles per litre) × volume (litres)
= 0.5 × 0.1 = 0.05 moles

4 Give the name, formula and number of moles of precipitate formed between each pair of solutions.

- 50 mL of 0.1 mol L⁻¹ silver nitrate (AgNO₃) is mixed with an excess of barium chloride (BaCl₂).
- 100 mL of 2 mol L⁻¹ sodium hydroxide (NaOH) is mixed with an excess of a solution of iron (II) sulfate (FeSO₄).

Calculate the mass of precipitate

EXAMPLE

iii 40 mL of 0.33 mol L⁻¹ potassium bromide (KBr) solution was added to an excess of lead (II) nitrate (Pb(NO₃)₂) solution. Give the name, formula and mass of the precipitate formed.

- The possible new compounds formed are potassium nitrate (soluble) and lead (II) bromide (insoluble); hence a precipitate of lead (II) bromide (PbBr₂) is formed.
- Net ionic equation: Pb²⁺ (aq) + 2 Br⁻ (aq) → PbBr₂ (s)
- Stoichiometry of reaction indicates that the ratio of moles of potassium bromide present to moles of lead bromide formed is 1:2.
(Stoichiometry deals with the relationships between quantities of reactants and products.)
- Calculating moles of potassium bromide present: 40 ÷ 1000 × 0.33 = 0.0132 moles. This means that there are 0.0066 moles of lead bromide formed.
- Mass of lead bromide = molar mass × number of moles
- Molar mass of lead bromide = (Pb × 1) + (Br × 2) = (207.2 × 1) + (79.9 × 2) = 207.2 + 159.8 = 367.0 g
- Mass of lead bromide = 367.0 × 0.0066 = 2.422 g

5 Give the name, formula and mass of precipitate formed between each pair of solutions.

- 40 mL of 0.5 mol L⁻¹ sodium carbonate (Na₂CO₃) solution mixed with excess magnesium sulfate (MgSO₄) solution.
- 80 mL of 0.2 mol L⁻¹ potassium sulfate (K₂SO₄) solution mixed with excess barium chloride (BaCl₂) solution.

6 Define each term.

- anion
- cation
- precipitate

Water is very important because of its physical and chemical properties, which result from the structure of its molecules. Water has similarities to ammonia (NH_3) and hydrogen sulfide (H_2S). Water and hydrogen sulfide molecules have 'V-shaped' structures. Ammonia is a triangular pyramid. Figure 1 shows the structures of these molecules.

- 1 Use figure 1 to help you draw Lewis electron dot structures of water, ammonia and hydrogen sulfide. Remember that Lewis electron dot structures show only outer (valence) electrons.

Electrons in a covalent bond can be distributed inequally between two atoms, due to differences in electronegativity (one end is slightly positively charged while the other is slightly negatively charged). When this happens, a polar molecule is formed. Water has polar covalent bonding in its molecules. The part of the molecule that is relatively negative is shown as δ^- and the relatively positive part as δ^+ . Figure 2 shows the polarity in a water molecule.

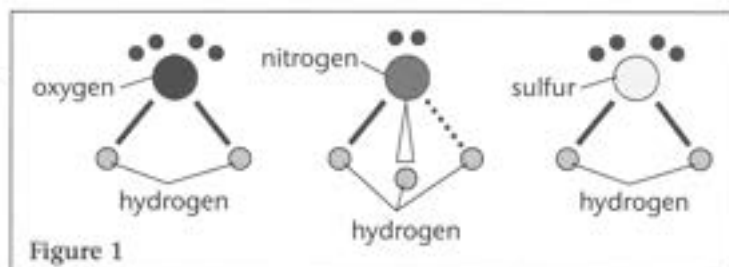


Figure 1

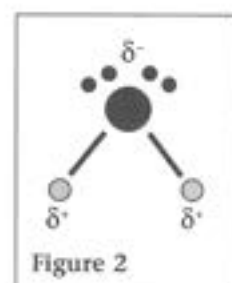


Figure 2

- 2 Which part of the water molecule is relatively positive?
- 3 Which part of the water molecule is relatively negative?

The dipoles formed (δ^+ and δ^-) attract dipoles in other molecules. This affects the physical properties of the compound. In water, there is a particularly strong form of dipole-dipole attraction called **hydrogen bonding**.

- 4 Define *polarity* and *dipole*.
- 5 Draw a diagram to show hydrogen bonding between two water molecules.

Boiling points of various hydrides

	Compound Group IV	Boiling point (K)	Compound Group V	Boiling point (K)	Compound Group VI	Boiling point (K)	Compound Group VII	Boiling point (K)
Period 2	CH_4	111	NH_3	310	H_2O	373	HF	292
Period 3	SiH_4	161	PH_3	188	H_2S	211	HCl	188
Period 4	GeH_4	185	AsH_3	216	H_2Se	232	HBr	206
Period 5	SnH_4	221	SbH_3	273	H_2Te	271	HI	238

Table 1

- 6 Use graph paper and refer to table 1.
 - a Draw your own set of axes as shown in figure 3.
 - b Plot the boiling point (K) of each hydride from table 1 on the y-axis of your graph.
 - c Label each hydride.
 - d Plot Period on the x-axis. (Period 2 is CH_4 to HF , Period 3 is SiH_4 to HCl , etc.)
 - e Join the boiling points of hydrides in the same Group.
- 7 Polar molecules have strong intermolecular forces (hydrogen bonds) that result in higher than expected melting and boiling points. Name three compounds from table 1 that you think have hydrogen bonds present.
- 8 Give the formula for:
 - a ammonia
 - b hydrogen sulfide
- 9 What are *valence* electrons?

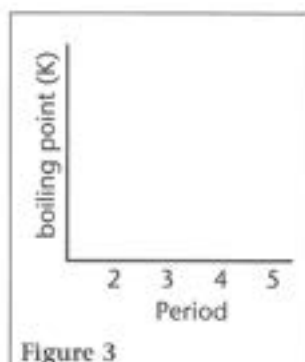


Figure 3

The number of electrons in the outer shell (valency) of an atom of an element affects the number and type of bonds this element will form with other elements. The electronegativity of an atom within a compound also affects the types of bonds formed. Electronegativity describes the attraction of an atom for electrons.

Electronegativity is the power of an atom to attract electrons.

- 1 Use a textbook to give another definition of *electronegativity*.

Non-metals with a strong 'desire' to gain electrons and complete their outer electron shells have high values of electronegativity. Metals with a strong 'desire' to lose electrons to form full outer shells have low values of electronegativity.

- 2 Explain how a metal can lose electrons to form a full outer shell (octet).

Generally, small atoms, especially those with nearly-full outer shells of electrons, attract electrons more easily than larger atoms do. Hence small atoms tend to have higher electronegativity values. The following generalisations can be made about bonding between atoms:

- When there is little or no difference in electronegativities a covalent bond is formed
- When there is a moderate difference in electronegativities a polar covalent bond is formed
- When there is a considerable difference in electronegativities an ionic bond is formed.

Figure 1 represents how electrons are 'shared' in bonds. The greater the size of the shaded area, the greater the share an atom has of a bonding electron. In polar covalent bonding the δ (delta) sign means **relatively**. Hence one end of the molecule is relatively positive ($\delta+$) to the other end ($\delta-$).



Table 1 shows the symbol (a), atomic number (b) and electronegativity (c) of some elements.

a	H	Li	Be	B	C	N	O	F	Na	Mg	Al	Si	P	S	Cl	K	Ca
b	1	3	4	5	6	7	8	9	11	12	13	14	15	16	17	19	20
c	2.1	1.0	1.5	2.0	2.5	3.0	3.5	4.0	0.9	1.2	1.5	1.8	2.1	2.5	3.0	0.8	1.0

Table 1

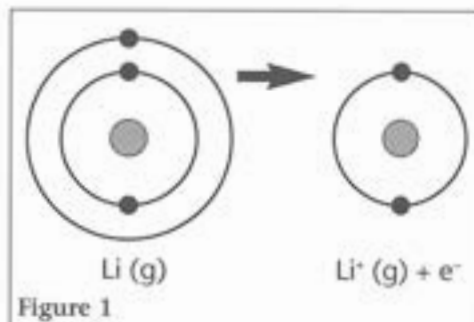
- 3 Refer to table 1 to answer the following.
 - a The values for helium, neon and argon are not quoted in table 1. Suggest why.
 - b Plot a graph of atomic number (x-axis) versus electronegativity (y-axis).
 - c What happens to the electronegativity values as you go across a Period?
 - d What happens to the electronegativity values as you go down a Group?
- 4 Use the electronegativities in table 1 to predict whether the compounds formed between the following elements would be covalent, polar covalent or ionic.
 - a potassium and chlorine
 - b carbon and sulfur
 - c hydrogen and water
- 5 Mark each statement as true or false.
 - a The most electronegative elements are the reactive non-metals in the top left-hand of the Periodic Table.
 - b The symbol δ means 'a positive charge'.
 - c Non-metals tend to form ionic compounds with metals.

Properties or characteristics of elements change in an ordered manner moving across the Periodic Table.

Ionisation energy

Scientists obtained information about how electrons are arranged in atoms by studying how much energy is required to remove electrons from these atoms. When electrons (negative charge) are removed from atoms (neutral charge), positive ions are formed. Figure 1 shows how a gaseous lithium atom forms a positive ion when an electron is removed.

The term **first ionisation energy of an element** refers to the energy required to remove one electron from each atom in a mole of gaseous atoms.



- 1 Why are positive ions formed when electrons are removed from electrically-neutral atoms?

Figure 2 shows a graph of ionisation energy (y-axis) against atomic number (x-axis) for the first 20 elements in the Periodic Table.

- 2 Give the name and symbol of each element.

- a atomic number 2
- b atomic number 10
- c atomic number 18

- 3 In which Group is each element in question 2?

- 4 Give the name and symbol of each element.

- a atomic number 3
- b atomic number 11
- c atomic number 19

- 5 In which Group is each element in question 4?

- 6 Give the name and symbol of each element.

- a atomic number 9 b atomic number 17

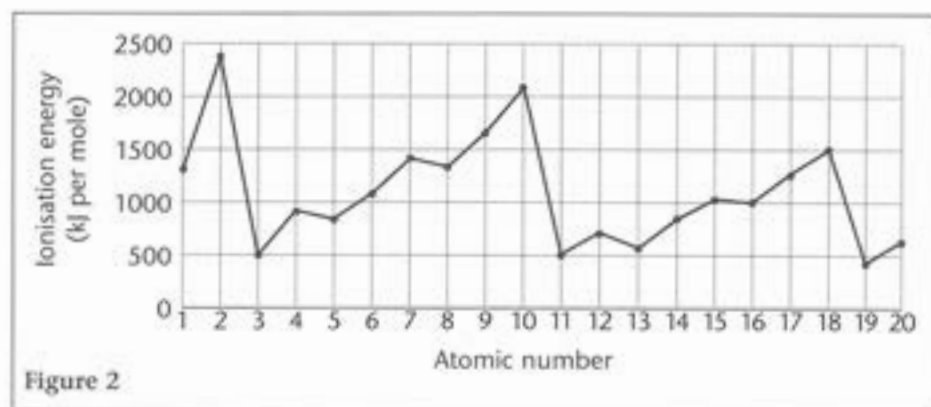
- 7 In which Group is each element in question 6?

- 8 Which element in figure 2 has the highest ionisation energy?

- 9 Which element figure 2 has the lowest ionisation energy?

- 10 Describe what happens to the ionisation energies of elements as you move across a Period.

- 11 Describe what happens to the ionisation energies of elements as you move down a Group.



The large ionisation energies of the noble gas (Group VIII) elements indicate that they are unreactive and have stable electronic structures (i.e. full outer shells, or octets).

- 12 What do you think a low ionisation energy indicates about a particular element?

- 13 Mark each statement as true or false.

- a It takes energy to remove an electron from an atom.
- b Noble gas atoms lose electrons easily.
- c Electrons have a negative electrical charge.
- d There are eight electrons in the first shell of an atom.
- e Ionisation energies are an example of a Periodic property
- f Removing electrons from an atom produces a negative ion.
- g Ionisation energy is the energy used to remove electrons from gaseous atoms.
- h An atom's ionisation energy can be used to suggest properties which that atom may possess.

Periodic properties are those that change in an ordered manner, progressing through the Periodic Table. The size of atoms is an example of a periodic property. Note that the size of an atom can vary due to outside effects on the electron cloud. Figure 1 shows the average proportional sizes of atoms for part of the Periodic Table.

- 1 Name two Groups of elements that are missing from figure 1.
- 2 Describe what happens to the size of an atom, in general, moving:
 - a left to right across a Period
 - b down a Group.

As you move across a Period from left to right, extra protons are being added to the nucleus and extra electrons to the outer electron shell. The increased attraction between the positive nucleus and electrons causes the electrons to be pulled closer to the nucleus.

- 3 Explain why atomic sizes increase moving down a Group.
- 4 Use the values in table 1 to draw a graph of atomic number (x-axis) against melting point (y-axis). Note that temperatures are in Kelvins (K). To change to degrees Celsius ($^{\circ}\text{C}$), add 273 to the Kelvin figure. The melting point quoted for carbon is its sublimation temperature.

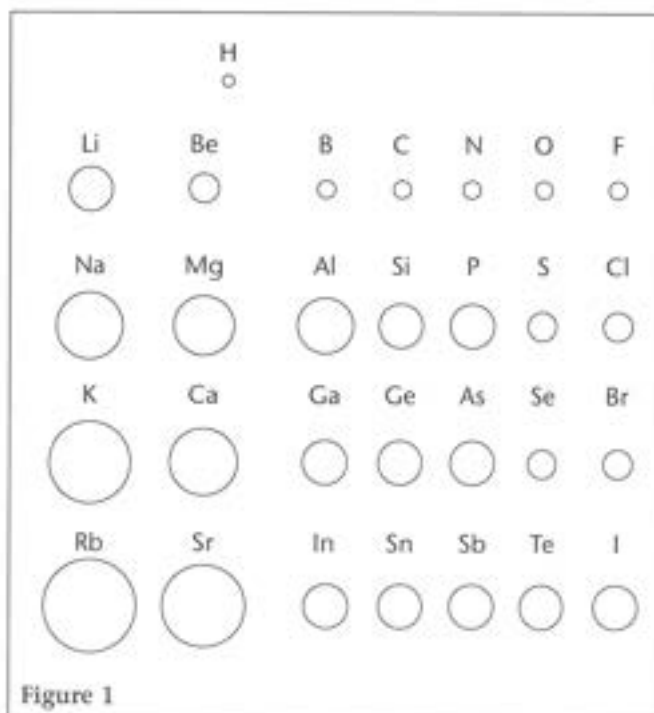


Figure 1

Element	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al
Melting point (K)	204	4	454	1551	2573	3573	77	55	53	27	371	924	933
Atomic no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Element	Si	P	S	Cl	Ar	K	Ca	Ga	Ge	As	Se	Br	Kr
Melting point (K)	1683	317	390	239	155	337	1115	303	960	304	490	266	116
Atomic no.	14	15	16	17	18	19	20	31	32	33	34	35	36

Table 1

Although there are some discontinuities (interruptions to the pattern), some trends should be visible.

- 5 Describe what happens to melting points moving down the elements in:
 - a Group II
 - b Group IV
 - c the noble gases.

The type of bonding present in the elements influences the structure and hence the melting points of elements. Major structures include metallic, covalent lattice (macromolecular) and discrete molecular (single, separate molecules). Metallic bonding is usually associated with high melting points.

- 6 Apart from a high melting point, what other physical property could be associated with a typical metal?
- 7 Carbon forms macromolecules (e.g. graphite and diamond). Name one other element that is likely to form a similar structure. Give reasons for your answer.
- 8 Which Group of elements exist as discrete atoms?
- 9 Explain why atomic sizes decrease from left to right across a Period.
- 10 Name the element that has the lowest melting point.
- 11 Name the element that has the highest melting point.

In chemistry you need to be able to:

- design and carry out first-hand investigations
- discuss first-hand investigations
- collect information from secondary sources
- assess information from secondary sources
- decide on the reliability of the results you obtain from your work
- decide on the reliability of any information your results suggest.

First-hand investigations are those that you carry out yourself, e.g. practicals in the laboratory. Secondary sources include textbooks and encyclopaedias.

- 1 Name two other sources of secondary information.
- 2 Many organisations do not allow people to begin scientific investigations until they have checked existing (secondary) information. Explain why.
- 3 Name two sources of secondary information you could use for designing and carrying out investigations on each topic.
 - a Which fertiliser works best on a particular crop?
 - b Which catalyst best speeds up a particular reaction?
- 4 Many people rely on recommendation or 'word of mouth' for information. How do these sources differ from other secondary sources of information?

Variables

During experiments, we often concentrate on observing and recording what happens when conditions change. Factors that can be changed during an experiment are called **variables**. Usually only one variable is changed at a time during an experiment. There are four major groups of variables.

- independent
- dependent
- controlled
- uncontrolled.

The **independent** variable is the major part of the hypothesis (belief) that you are testing. It is the condition that you change to see what happens. The **dependent** variable is the feature that changes as a direct result of changing the independent variable. A **controlled** variable is one that is kept the same in each experiment. An **uncontrolled** variable is one that has not been taken into account during the planning of the experiment.

- 5 A student noted that when zinc metal was added to dilute hydrochloric acid, hydrogen gas was given off. She observed that the smaller lumps of the metal seemed to react quicker than the larger lumps. She added 50.0 mL of 1.0 mol L⁻¹ of hydrochloric acid to 5.00 g of zinc metal and measured how long it took for a volume of gas to be given off. She then repeated the experiment using the same mass of zinc but this time it was in powder form.

Hypothesis	Variables		
	Independent	Dependent	Controlled
Smaller particles react quicker than larger lumps	Size of zinc	Time for reaction to occur	Volume of acid Concentration of acid Mass of zinc

Table 1

- a Study table 1, which summarises the variables in this experiment.
- b Identify an uncontrolled variable in the experiment.

6 A student wanted to test if increasing the concentration of acid increased the rate of evolution of gas in the reaction between zinc and hydrochloric acid. She added 5.00 g of zinc metal to 50 mL of acid and timed how long it took to evolve a fixed mass of gas. She repeated this, using different concentrations of acid. Her results are shown in table 2.

Using the format shown in table 1, identify the hypothesis, independent, dependent and controlled variables for this experiment.

Time (sec)	75	30	20	15	12
Concentration (mol L ⁻¹)	0.1	0.5	1.0	1.5	2.0

Table 2

Three terms are very important when it comes to assessing data:

- accuracy
- reliability
- validity

Accuracy: does not contain any errors

The mechanic was correct, he made an accurate diagnosis of the fault in the car.

Reliability: dependability; giving the same result every time the test is carried out

The indicator was reliable – every time it was added to an acid it changed from pink to colourless.

Validity: how relevant and appropriate a process is

She was convinced of the validity of her method; it did measure what it was supposed to measure.

- 7 Give another definition of *accuracy*.
- 8 Write a sentence illustrating the meaning of the word *accuracy* in a scientific context.
- 9 Discuss this statement: *A stopped clock is more accurate than one that is always exactly five minutes late.*
- 10 Sometimes results can be accurate to $\pm 5\%$ or $\pm 10\%$.
 - a Give an example of when this level of accuracy (or slight inaccuracy) would be acceptable.
 - b Give an example of when 100% accuracy is required.
- 11 Give another definition of *reliability*.
- 12 Write another sentence illustrating the meaning of the word *reliability* in a scientific context.
- 13 Discuss this statement: *He set out for the bus, knowing he could rely on it being 5 minutes late.*

In science, reliability is discussed in terms of anyone using a particular method or following the same set of instructions getting the same results. These results might be incorrect if the method is flawed or the apparatus is not designed for that experiment. Data collection methods must be valid.

- 14 Give another definition of *validity*.
- 15 Write another sentence illustrating the meaning of the word *validity* in a scientific context.
- 16 Discuss this statement: *When he tried to board the plane they questioned the validity of his passport.*
- 17 A group of scientists declared that their investigations were reliable and that the conclusion they drew about a given drug trial was valid. A second group of scientists disputed the statement, claiming that the conclusions drawn were unreliable and therefore the conclusion was invalid. Given that both groups of scientists were competent, suggest why this type of dispute could occur.
- 18 Mark each statement as true or false.
 - a Dependent variables can change as a direct result of altering independent variables.
 - b A valid result is based on well-supported facts.
 - c All scientific results are valid if they are accurate to $\pm 5\%$ or $\pm 10\%$.
 - d Controlled variables remain the same in each experiment.

Metals are found on the left-hand side and centre of the Periodic Table. Most metals are lustrous (shiny) and they are often silver in colour. Metals are good thermal and electrical conductors, whereas most non-metals are poor conductors.

1 What is meant by a *good thermal conductor*?

Metals are malleable; they can be beaten into shape. They are also ductile; they can be drawn into wires. Non-metals are brittle; they snap instead of bend. Metals are usually denser than non-metals. Metals are solids at room temperature (with the exception of mercury, Hg). They usually have high melting points. Very few metals occur **native** (uncombined) in the natural state. They are usually found combined with other elements. The earliest metals to be used by humans were copper (Cu), tin (Sn), silver (Ag) and gold (Au) – all of these metals can be found native.

2 Explain why copper, tin, silver and gold can be found native.

Very few metals occur uncombined in the natural state. They are usually found chemically joined with other elements in **minerals**. A mineral is a naturally-occurring pure substance. An ore is a material from the Earth's crust that contains enough of a substance to make it worth mining. Iron (Fe) exists in the forms of haematite and magnetite, which are both forms of iron oxide (i.e. iron combined with oxygen in a compound). Iron ore is very similar in chemical composition to iron rust.

Aluminium (Al) is found as bauxite (aluminium oxide) and lead (Pb) as galena (lead sulfide, lead combined with sulfur). Copper occurs in the ores malachite and azurite, which are both forms of copper carbonate. Another important ore of copper is chalcopyrite, a yellow mineral consisting of sulfides of copper and iron.

3 Define each term.

- | | |
|-------------|-----------|
| a lustrous | d brittle |
| b malleable | e ore |
| c ductile | f mineral |

4 Give the chemical name for each substance.

- | | |
|-------------|-------------|
| a haematite | d limestone |
| b bauxite | e malachite |
| c galena | f quicklime |

5 Give the names and symbols of the elements present in each substance.

- | | |
|-------------------|---------------------|
| a iron oxide | d copper carbonate |
| b aluminium oxide | e hydrogen peroxide |
| c lead sulphide | f ammonia |

6 Explain why ores are non-renewable resources.

Mining

Lead was the first metal mined in Australia, in Glen Osmond (South Australia) in 1841. The mining and minerals industry in Australia is very important to the economy. It employs about 200 000 people and is indirectly responsible for another 300 000 jobs. It creates employment and brings in valuable foreign currency. However, despite the economic advantages that mining brings, the mining and minerals industry does cause problems.

7 Define *mining*.

8 Outline some possible negative aspects to mining.

9 Name three substances that are mined in Australia.

10 Outline two benefits of mining.

11 Often *raw* materials are mined. What does *raw* mean in this context?

In order to extract metals from their ores, energy is needed. The more reactive the metal, the more difficult it is to decompose (break down) its ore.

Copper The most important ores, the sulfides, contain between 1 and 12 percent of copper. Ores are mined and then crushed and concentrated by **flotation**. This technique allows comparatively heavy substances to float on water. Finely-ground copper ore is mixed with water and flotation reagents are added. Air is blown through this mixture to form a froth. The froth mixes with the sulfide but not with the other substances. The ore floats to the top, leaving other substances behind. Oxides and carbonates of copper are reduced (oxygen, etc. removed) with carbon. Sulfides are smelted in a furnace, resulting in copper that is approximately 98 percent pure. This can be further purified to 99.9 percent through electrolysis. Electrolysis uses large amounts of electricity to attract pure copper to an electrode.

1 Define each term.

a energy b sulfide c flotation d reagent e reactive

2 Draw a flow chart to show the production of copper metal from its ore.

Iron is extracted from iron ore (haematite) in a blast furnace. Coke (a form of carbon) and limestone (calcium carbonate) are added with iron ore to a blast furnace where the raw materials are strongly heated. The coke provides the heat to produce carbon monoxide (CO) which takes the oxygen from the iron oxide. The limestone removes impurities, which melt in the furnace and are removed as slag. Hot air is blown in near the bottom of the furnace. The molten iron produced is then cast into ingots. This cast iron usually requires further treatment before it can be used.

3 Suggest why it is important to remove impurities from substances being mined.

4 Draw a flow chart to show the production of iron metal from its ore.

More active metals such as aluminium (Al) need to be extracted from their ores by electrolysis. Sodium is extracted from sodium chloride (NaCl, rock salt) by this method.

5 Write balanced formula equations for each reaction.

a copper (I) sulfide + oxygen → copper + sulfur dioxide

b iron (III) oxide + carbon monoxide → iron + carbon dioxide

c sodium chloride → sodium + chlorine

d aluminium oxide → aluminium + oxygen

The Bronze Age This was the period of ancient human culture 4000–3000 BC. It was characterised by the human use of bronze. This age ended when the Iron Age began. The Greeks and Roman used bronze to make tools, weapons, coins and objects of art. Bronze is an alloy, usually containing copper and tin.

6 Define alloy.

The Iron Age This was a period in human culture when iron was commonly used for making tools and weapons. (Note that bronze was replaced by iron at different times in different cultures.) The first known use of iron was in Egypt (c. 4000 BC). Modern iron processing began in Europe in the 14th century.

7 Discuss why bronze was used by humans before iron was.

8 Match each term to the correct definition below:

alloy, electrolysis, furnace, haematite, ingot, ore, reduction, slag, smelt

a Heating an ore with a chemical change to obtain the metal: _____

b A piece of metal cast into a shape: _____

c A waste product of steel production: _____

d A mixture of metals: _____

e Passing an electrical current through a substance to cause a chemical reaction: _____

f A hard, red oxide of iron: _____

g A structure that uses heat to extract metals from their ores: _____

h The chemical removal of oxygen from a substance: _____

i A mineral that contains enough of a metal to be worth mining: _____

Reactivity of metals

Some metals are more reactive than others. Most (active) metals can be made to react with oxygen, water, acids and alkalis. Metals such as gold (Au) and silver (Ag) are very unreactive.

Metal + Oxygen → Metal oxide

1 Write a balanced formula equation for each reaction.

- a Copper + oxygen → copper (II) oxide
- b Iron + oxygen → iron (III) oxide

Metal + Water → Metal hydroxide + Hydrogen

2 Write a balanced formula equation for each reaction.

- a Sodium + water → sodium hydroxide + hydrogen
- b Potassium + water → potassium hydroxide + hydrogen
- c Calcium + water → calcium hydroxide + hydrogen

Metal + Acid → Salt + Hydrogen

Metals such as magnesium (Mg), zinc (Zn) and aluminium (Al) react with acids.

3 Write a balanced formula equation for each reaction.

- a Zinc + hydrochloric acid → zinc chloride + hydrogen
- b Magnesium + hydrochloric acid → magnesium chloride + hydrogen
- c Magnesium + sulfuric acid → magnesium sulfate + hydrogen

Recycling metals

Our supply of metals or minerals is limited by how much of them is in the ground and how much of that is obtainable. Since 1950, more metals have been used up than in the whole history of the world before 1950. It is difficult to estimate how long our supply of metals will last, so we should be more careful about how much metal we waste. As metals become more expensive, recycling becomes more attractive. Producing copper from scrap costs about 4% of the cost of making it from its ore. Today about 50% of copper and about 80% of the lead from car batteries is recycled. About 20% of aluminium and 18% of iron (as steel) is recycled.

- 4 Use the example of aluminium drink cans to outline the steps in the process of recycling aluminium. Discuss whether recycling aluminium requires the same amount of energy as extracting it from ore.
- 5 Outline reasons, other than cost, why containers should be recycled.
- 6 Outline why aluminium is a good material for making drink containers.

Metal alloys

Metal alloys are mixtures of two or more metals. They take advantage of the properties of a number of metals in order to produce properties that are superior to those of pure metals. Bronze is an alloy of copper and tin. It is hard and does not corrode. Cupronickel is 75% copper and 25% nickel. It is an attractive silver in colour and is hardwearing. Steel is an alloy of iron with some carbon added. There are many types of steel and all of these contain iron, but the other metals present vary. Brass contains roughly 70% copper and 30% zinc. It dully in appearance but doesn't corrode. Duraluminium contains 94.5% aluminium, 4% copper, 1% manganese and 0.05% magnesium. This alloy is three times as strong as steel, but only one-third as dense. Solder is 70% tin and 30% lead. It is soft and has a low melting point.

- 7 Why are alloys used instead of pure metals?
- 8 Draw up a table to present the information on alloys above. Include the metals that make up alloys, their properties and possible uses.

Write the letter of each definition beside the correct term.

Term	
1 alloy	
2 reactivity	
3 aluminium	
4 steel	
5 bauxite	
6 recycle	
7 blast furnace	
8 ore	
9 bronze	
10 metal	
11 coke	
12 iron	
13 compound	
14 haematite	
15 corrosion	
16 atom	
17 limestone	
18 sulfide	
19 hydrogen	
20 lead	
21 malleable	
22 lustrous	
23 ductile	
24 abundance	

Definition
a Common metal with the symbol Fe.
b An alloy containing mainly copper and tin.
c A form of calcium carbonate used in a blast furnace.
d Describes how a metal behaves; e.g. with an acid.
e Mineral from which a metal can be extracted.
f Element on left-hand side of Periodic Table.
g Shiny.
h How much there is of a substance.
i Can be drawn into wires.
j An alloy of iron.
k Can be hammered into shape.
l To use again.
m Compound containing sulfur.
n A mixture of two or more metals.
o Aluminium ore.
p Soft, dense metal that doesn't corrode easily.
q Lightweight metal used to make windows and doors.
r Gas given off when metals react with acids.
s Simplest particle of an element.
t Machinery used to produce iron.
u Iron ore.
v 'Rust'.
w Two or more different elements combined.
x Form of carbon used in a blast furnace.

Research

25 The extraction of metal ores is a very important industry for Australia. Draw up a table showing:

- Name of metal
- Ore from which it is extracted
- A location where it is mined
- The approximate cost of this metal.

Complete the table for four different metals.

The **catchment** area is the area from which rainfall and water flow is collected for use by a community.

- 1 Describe (or provide a map of) your local catchment area.
- 2 Name some organisations that you could contact to find out more information about your local water supply.
- 3 Suggest reasons why catchment areas are important.

Contamination occurs when foreign matter gets into water supplies. This lowers the quality of the water and, in severe cases, can make the water unfit for human use.

- 4 Outline some possible types of contamination in your local catchment area and the sources of these contaminants.
- 5 There are many chemical tests available for identifying what contaminants are present and the amounts of these substances. Study the list below and outline how each could be identified and measured.
 - a Heavy metals
 - b Common ions (Name two examples.)
 - c Acidity
 - d Hardness
- 6 Suggest why some contaminants can be dangerous in small amounts while others only cause problems in large amounts.

Water is treated before it is supplied to the mains. Each water authority has its own treatment procedure dependent upon the initial quality of water, but the following procedure is fairly typical.

Water from a dam passes through large mesh filters to trap any large objects. The water is clarified (made clearer) by precipitating out some impurities. Alum and iron (III) hydroxide are used as precipitating (flocculating) agents. The larger suspended particles (floculate) are filtered out, often by passing through a bed of sand. The water is then aerated by cascading through waterfalls or in artificial fountains. This removes odours and tastes by oxidising them. The water can be softened (calcium and magnesium ions removed) by the addition of lime (CaO) and soda (Na_2CO_3). At this stage there may still be microorganisms present so these have to be killed. Chlorine is added (around 1 to 2 ppm) to sterilise the water. Some very resistant microorganisms may survive this process. In some cases ozone (O_3) is used instead of chlorine. Ozone decomposes to form oxygen and is regarded as less of an environmental concern than chlorine. Many areas also add fluoride, which helps protect people's teeth from decay.

- 7 Define the *mains* in the context of water.
- 8 Suggest how a bed of sand could be useful in filtering.
- 9 Draw a flow chart to show the sequence of steps in a typical purification process.
- 10 The name *alum* can refer to many different chemicals. One alum used in water treatment has the formula $\text{AlK}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$. Name this compound.
- 11 Give the formula for ozone.
- 12 Explain why ozone is used in preference to chlorine.
- 13 Why is fluoride added to water supplies?
- 14 Give the full chemical name for lime.
- 15 Give the full chemical name for soda.
- 16 Read the following passage and define each word in *italics*.

Microscopic *membrane* filters are used to *eliminate* almost all small *pollutants* in water. They are composed of artificial *polymers* (often polycarbonate), usually consisting of fibres the size of a human hair. The fibres have microscopic holes to filter out impurities, allowing only water to pass through. The holes are smaller than one *micron* (10^{-6} metres) and compared to other processes this process provides improved protection from water-borne *pathogens* such as *Giardia* and *Cryptosporidium*.

When soap solution is added to soft water and shaken, a lather (foam) readily forms. If soap solution is added to hard water, lather does not form immediately. More soap has to be added until it removes all of the hardness. Hardness in water wastes soap. Other indications of hard water include hard deposits in hot water systems and kettles. Synthetic ('soapless') detergents are not affected by hard water.

There are two sorts of hardness: temporary and permanent. Temporary hardness is caused by carbonates and can be removed by boiling. Permanent hardness is not removed by boiling. Chemicals called *water softeners* can be used to remove hardness so that less soap is needed. Dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions are a major source of hardness. Water can also be 'softened' by passing it through an ion exchange medium, where calcium and magnesium ions are replaced by sodium ions. The medium is usually made up of organic resins and can be regenerated by passing brine (salt solution) through it. Chemical water softeners work in a different way; they react with these ions and remove them from solution by precipitating them out.

A student carried out a series of tests to compare two water softeners: washing soda (sodium carbonate decahydrate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) and a commercial brand, *Softa*. Below is an extract from her practical report.

Method 50 mL of hard water was placed into a conical flask. 1.0 mL of soap solution was added to the flask, which was then stoppered and shaken. More soap solution was added, 1.0 mL at a time and shaking repeated until a permanent lather formed. The total volume of soap solution used was recorded.

This procedure to obtain a lather was repeated, but 0.1 g of sodium carbonate was added first. This was done with other quantities of sodium carbonate and also with equivalent amounts of *Softa*.

Results

Mass of softener (g)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7
Volume of soap using sodium carbonate (mL)	24.0	16.0	8.0	3.0	3.0	3.0	3.0	3.0
Volume of soap using <i>Softa</i> (mL)	24.0	19.0	14.0	9.0	5.0	1.0	1.0	1.0

- Name the piece of apparatus that should be used to measure out each of the following.
 - the hard water sample
 - the soap solution
- Outline how the student ensured that this procedure was a 'fair test'.
- On the same set of axes, draw a graph of volume of soap solution (y-axis) against mass of softener (x-axis) for each softener.
- Referring to your graph from Q3, discuss the relative effectiveness of the two chemicals as water softeners.
- Name one way of avoiding the problems of using soap rather than softening the water.
- Describe some indications that a domestic water system has hard water.
- Outline the difference between temporary and permanent hardness.
- When ions from two soluble compounds react to form an insoluble substance. Which chemical phenomenon does this describe?
- Distillation can be used to remove hardness from water but is not used on a large scale. Give a reason why distillation isn't used to purify water in large quantities.
- What is *brine*?

- Research**
- Find out the actual quantities of dissolved solids required for water to be classified as *hard*.
 - Hard water does have some advantages. Find out two of these advantages.

It is important that the water supplied for home use, business and agriculture is of a good quality and free from pollutants. Some tests on water quality can be carried out in the school laboratory. One indication of water quality is the amount of other substances in the water. These materials can be suspended or dissolved.

Turbidity

Suspended material in water will make it cloudy. The small, floating particles scatter light. The amount of scattering depends on the size and concentration of the particles. Figure 1 shows two ways of measuring turbidity (apparatus not to scale). The first apparatus consists of a glass tube that has a cross on its base which can be seen by looking down from the top of the tube (not visible in figure 1). There is a scale of depth on the side of the tube. Water is poured into the open top until the cross at the base is no longer visible.

The second piece of equipment is a patterned disc attached to a cord which is marked at regular distances. The disc is placed into the water and lowered under water until the pattern on the disc can no longer be seen.



Figure 1

- 1 Describe how the first apparatus could be used to compare the turbidity of a range of water samples in a school laboratory.

Figure 2 shows a way to measure the amount of light transmitted through a water sample.

- 2 Describe how the apparatus in figure 2 could be used to compare the turbidity of water samples.
- 3 Discuss the accuracy of the results obtained from the apparatus in figure 1 in comparison with the apparatus in figure 2.

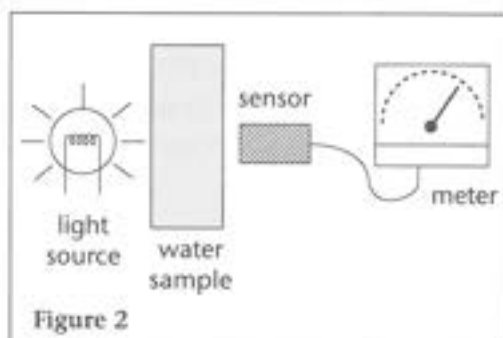


Figure 2

Total dissolved solids (TDS)

This is the total mass of solids dissolved in a known quantity. TDS is often quoted in parts per million (ppm). There are two main ways of measuring the amount of dissolved solids: evaporation and using electrical conductivity.

A student carried out a series of experiments to investigate water quality. Extracts from his report are shown below.

Evaporating water samples

The mass of an empty container was found. 100.00 mL of water sample was placed in the container and the water evaporated. The container was re-weighed. The mass of solid remaining (the residue) was calculated. This was done with five samples of water.

	Mass of empty container (g)	Mass of container + residue (g)	Mass of residue (g)
Tap water	57.51	57.76	
Bore water #1	58.36	58.69	
Bore water #2	58.02	58.82	
Sea water	56.41	59.76	
Distilled water	57.93	57.96	

Table 1

- 4 Name a piece of equipment that could be used to measure the volume of liquid accurately.
- 5 Calculate the masses of solids dissolved in the water samples.
- 6 State the assumptions you made in your answers to Q5.
- 7 Use your answers to Q5 to quote the amount of dissolved solids in the samples in ppm.

Electrical conductivity of water samples

Ions are charged particles that can carry a current in solution. The more ions there are in solution, the greater the current that can flow. This principle can be used to measure the amount of dissolved ionic solids in a sample of water.

A student used the apparatus shown in figure 3 to investigate the conductivity of some water samples. Extracts from her report are shown below.

Method

The apparatus shown in the diagram (figure 3) was set up. The apparatus was calibrated with standard potassium chloride solutions. The conductivity of the water samples was measured.

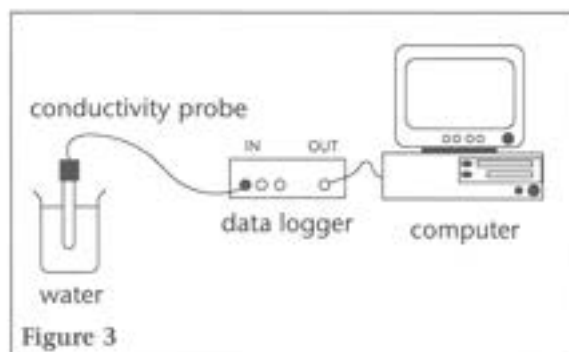
Results

Concentration KCl (ppm)	0	200	400	600	800	1000
Reading (arbitrary units)	20	290	550	820	1080	1350

Table 2

Sample	Distilled water	Tap water	Bore #1 water	Bore #2 water	Sea water
Reading (arbitrary units)	30	160	320	910	Off scale

Table 3



- 8 Use the data from table 2 to draw a 'best-fit' line graph showing the calibration of the apparatus. Put the conductivity reading on the y-axis and the concentration of potassium chloride on the x-axis.
- 9 Refer to your graph from Q8 to calculate the quantity of dissolved solids in each water sample.
 - a distilled water
 - b tap water
 - c bore #1 water
 - d bore #2 water
- 10 Give the formula for potassium chloride.
- 11 Outline the purpose of the data logger in this experiment.
- 12 Outline the purpose of the computer in this experiment.
- 13 Give a possible reason for why the reading for sea water was off the scale.
- 14 Outline the assumptions made in this method of finding total dissolved solids in a sample of water.
- 15 What is meant by *arbitrary units*?
- 16 What is meant by *calibrate*?
- 17 What is meant by *turbidity*?
- 18 Discuss the advantages and disadvantages of using electrical conductivity compared to evaporation of water samples to find the amount of total dissolved solids in a sample of water.

Research

- 19 Define *nephelometer*.
- 20 Find out the accepted TDS levels for *potable* (fit for human consumption) water in your area.
- 21 Chlorine is a poisonous substance yet it is commonly used in drinking water and swimming pools. Why is it used? Why doesn't it poison humans?

The quality of water is determined by many factors. Two important considerations are the amount of oxygen dissolved in the water sample and the **biological oxygen demand (BOD)** of that sample. BOD is an indication of how much oxygen is needed by organisms in a sample of water.

Oxygen becomes dissolved in water in many ways; including by diffusing from the atmosphere, by aeration (rapid movement) and by underwater green plants.

- 1 What is the name of the process by which green plants produce oxygen?
- 2 Where might aeration occur in a body of water?

The solubility of oxygen in water varies with temperature. Table 1 shows how much oxygen can dissolve in fresh water at various temperatures at 1 atmosphere pressure.

Temperature (°C)	1	4	8	12	16	20	24	28	32
Dissolved oxygen (ppm)	14.2	13.2	11.9	10.8	9.9	9.2	8.5	7.9	7.4

Table 1

- 3 Draw a 'best fit' line graph of dissolved oxygen (y-axis) against temperature (x-axis). Label the axes.
- 4 Describe the relationship between temperature and how much oxygen can dissolve.

The figures in table 1 relate to the maximum oxygen that can dissolve at a particular temperature. This does not mean that this quantity of oxygen will definitely be present.

- 5 A fish tank is 1.5 m long by 0.5 m wide by 0.5 m high. It is full to the brim with water. The tank is aerated and the water present is saturated with oxygen. Calculate the mass of oxygen dissolved in the tank at 20°C.

(Hints: 1 cm³ of water = 1 mL; 1 ppm = 1 mg L⁻¹)

Aerobic life forms are those that require oxygen for life. The amount of oxygen dissolved in water can be a very good indicator as to the health of a water ecosystem. Most natural water systems need at least 5 ppm of dissolved oxygen to support life. Because the amount of dissolved oxygen is such a good indicator of the health of a water-based ecosystem, it is important to measure the level of dissolved oxygen at regular intervals. This can be done with a titration method.

- 6 Define *titration*.

Another method for measuring dissolved oxygen uses an oxygen sensor connected to a data logger (see figure 1). The oxygen sensor is an electrolytic cell. The more oxygen there is in the water sample, the greater the flow of electricity. Low dissolved oxygen content can encourage organisms that use anaerobic fermentation. This process can release toxic gases into the water.

- 7 You have been asked to monitor the level of dissolved oxygen in a nearby stream over a period of time. Describe how you would do this and how you would ensure your results were valid.
- 8 Explain what is meant by an *anaerobic organism*.
- 9 A researcher found that the dissolved oxygen concentration in a body of water decreased with depth. Give a possible reason for this.
- 10 What is meant by *toxic*?

Research 11 What is meant by *electrolytic*?

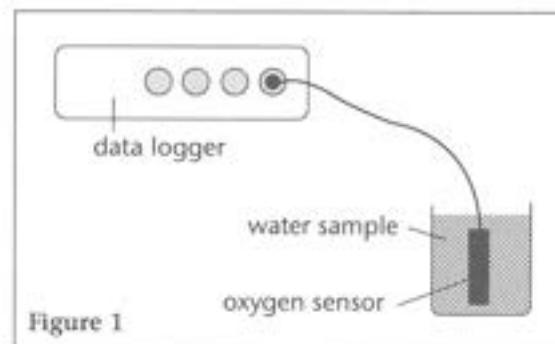


Figure 1

Biological (biochemical) oxygen demand is a very good indicator of the concentration of organic waste, bacteria and other aerobic organisms in a sample of water. The BOD of a sample of water can be determined in different ways. The following method, using the formula shown in figure 1, is a common one.

The biological oxygen demand (BOD) is the amount of dissolved oxygen consumed under specific conditions in a given time by the aerobic biological oxidation of organic (plant and animal) matter in a specific sample of water.

$$BOD = (D_1 - D_2) \div P$$

Figure 1

- Dilute the sample with distilled water (to ensure that not too much organic matter is present). Let the dilution factor = P .
- Measure the dissolved oxygen content. (D_1 , in mg L^{-1})
- Incubate the diluted sample for 5 days at 20°C .
- Measure the dissolved oxygen content again. (D_2 , in mg L^{-1})
- The BOD is then calculated by finding the difference between the dissolved oxygen contents and then dividing by the dilution factor, as shown in figure 1.

EXAMPLE

- A sample of waste water was taken. 1 mL of this sample was made up to 100 mL using pure water. The D_1 of the diluted sample was found to be 7 mg L^{-1} . After 5 days incubation at 20°C , the D_2 was found to be 3 mg L^{-1} . Calculate the BOD for the sample.

$$\begin{aligned} BOD &= (D_1 - D_2) \div P \\ &= (7 - 3) \div 0.01 \\ &= 400 \text{ mg L}^{-1} \end{aligned}$$

- A sample of waste water was taken. 1 mL of this sample was made up to 200 mL using pure water. The D_1 of the diluted sample was found to be 4 mg L^{-1} . After 5 days incubation at 20°C , the dissolved oxygen content was found to be 2 mg L^{-1} . Calculate the BOD for the sample. Show your working.

Samples are incubated for five days because historically this was supposed to be the longest time that river water takes to flow from source to sea in the United Kingdom. The more oxygen that is used up, the greater the concentration of organic waste. If the BOD is high, decay organisms can use up all of the available oxygen, which harms other aquatic organisms.

- In what ways can aquatic organisms be harmed if decay organisms use up the available oxygen?
- A measure of the mass of dissolved oxygen consumed in biological processes that degrade organic matter in surface water. What does this define?
- Figure 2 shows how the concentration of organic substances changes over time in the presence of organisms that consume organic material. Write an interpretation of what is shown in figure 2.
- Explain why testing BOD of water samples from the output of a water treatment plant could indicate how efficient the plant is.
- Write the missing word: Organic matter that can be decomposed by biological means is said to be _____.
- What is meant by *incubate*?

Research 8 Give your own definition of *biological oxygen demand*.

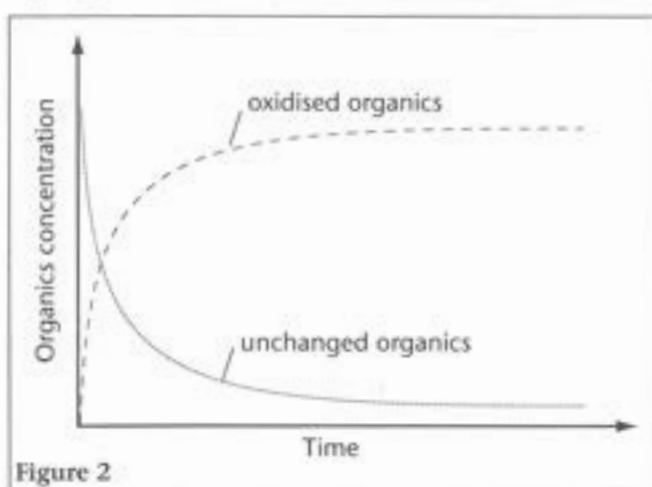


Figure 2

Viscosity can be defined as the resistance of a fluid to flowing or changing shape. The more viscous a substance is, the slower it flows. Viscosity has important industrial relevance, especially when fluids are being transported through pipes. A group of students was set the task of measuring the viscosity of liquids. The practical procedure was set out as follows:

- Set up a clean and dry burette with a beaker underneath as shown in figure 1
- Place liquid into the burette to slightly above the zero mark
- Run out liquid until the meniscus rests on the zero mark
- Time how long it takes for 50 mL of the liquid to pour out of the burette into the beaker
- Rinse out the burette with the next liquid to be tested
- Repeat with the other liquids.

Table 1 shows the results collected from the six groups who carried out the experiment.

Liquid	Water	Milk	Ethanol	Cooking oil
Time to flow (sec)	22, 24, 27, 45, 29, 23	29, 30, 34, 19, 35, 31	25, 27, 31, 55, 33, 28	66, 72, 80, 122, 90, 71



Figure 1

Table 1

- 1 Use a diagram to help you explain what a *meniscus* is.
- 2 Describe how a burette is used to measure out a volume of liquid.
- 3 One group had problems with a burette. From each set of results, pick out the figure that is probably due to the faulty burette.
- 4 Leaving out the faulty readings, work out the average time for 50 mL of each liquid to flow from a burette.
- 5 List the liquids in order of increasing viscosity.
- 6 Why should the burette be rinsed out between experiments?
- 7 Name any essential apparatus not shown in figure 1 that is required to carry out the experiment.

Fluid particles have to move past each other in order to flow; therefore viscosity can be regarded as being due to the friction between moving molecules. Polar (electrically-charged) molecules are more likely to interact with each other than similar neutral molecules are. Viscosity changes with temperature; liquids flow better at higher temperatures, gases flow better at lower temperatures.

- 8 Describe what happens to the viscosity of each substance.
 - a liquids as the temperature is decreased
 - b gases as the temperature is increased
- 9 The students were asked to devise an alternative way of comparing the viscosity of liquids. One student included a diagram (see figure 2) but didn't give any instructions for carrying out the experiment. Write a set of instructions for comparing viscosities of liquids using the apparatus shown in figure 2. Include details of how you would ensure it was a fair test.
- 10 Name a viscous liquid not already mentioned on this page.
- 11 Mark each statement as true or false.
 - a Fluids are liquids and gases.
 - b The greater the interaction between fluid particles, the less the viscosity of that fluid.
 - c The rate of flow of a fluid is inversely proportional to its viscosity.
 - d Liquids are more viscous than gases.
 - e Viscosity of fluids is related to temperature.
 - f Polar molecules don't attract each other.
 - g Blood is more viscous than water.
 - h Viscosity can be defined as a substance's resistance to flow.

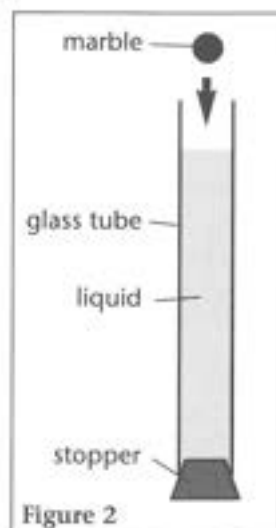
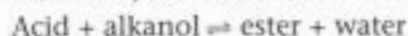


Figure 2

Saponification (literally 'making soap') is an important industrial process.

- 1 State one physical difference between a fat and an oil.

Fats and oils are types of **esters**. Figure 1 shows the general structure of an ester. R_1 and R_2 are different alkyl groups. Esters are formed by the reaction between an alkanolic (carboxylic) acid and an alkanol (alcohol):



- 2 What is the significance of the symbol ' $\rightleftharpoons$ ' in the equation above?

Fats and oils tend to be long chain triglycerides (*tri*, meaning three). Figure 2 shows the structures of glycerol and a typical triglyceride. The R is a long chain hydrocarbon group.

- 3 Give the systematic (IUPAC) name for glycerol.
- 4 Glyceryl tristearate is a typical animal fat. Using the structure shown in figure 2 and given that the R (stearate) group for glyceryl tristearate has 17 carbon atoms, draw the structure for this compound. Assume that the hydrocarbon group is saturated.

One version of the discovery of soap is that animal fats from meat being cooked over a wood fire dripped into the ashes. The ashes contained alkaline potassium carbonate that reacted with the fats to form an impure form of soap. The people who raked out the ashes then washed their hands in water found that the mixture cleaned their hands better than water alone did. Around 200 years ago, soap was made by boiling animal fats with alkaline solutions. A similar process is used today but generally vegetable oils are used and there are extra steps to ensure the purity of the finished product. A triglyceride is heated with sodium hydroxide solution forming glycerol and a sodium carboxylate (soap). Figure 3 shows an outline of this process. Glyceryl tripalmitate (a fat derived from palms) is commonly used as a raw material. It forms sodium palmitate.

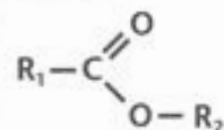


Figure 1

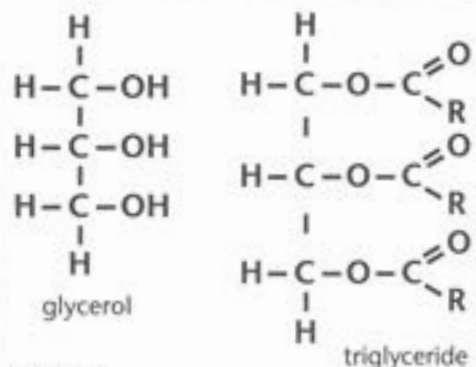


Figure 2

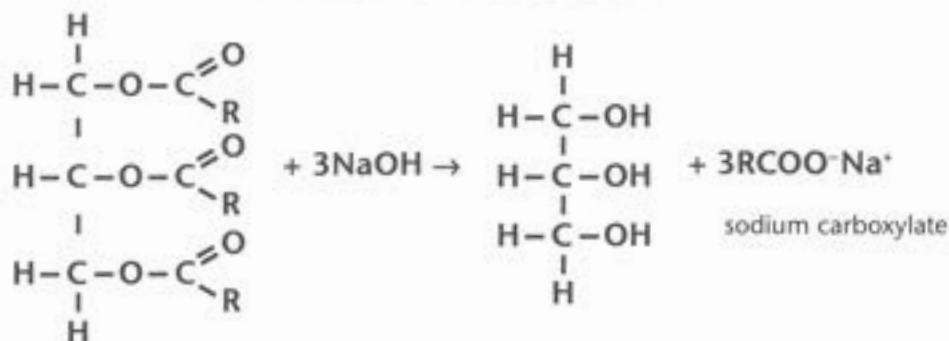


Figure 3

- 5 Using the structure shown in figure 3 and given that the R (palmitate) group for sodium palmitate has 15 carbon atoms, draw the structure for this compound. Assume that the hydrocarbon group is saturated.

The industrial production of soap can involve the breaking down of the triglyceride into fatty acids before it is treated with sodium hydroxide. Fatty acids do not occur naturally as separate compounds, but only bound with glycerol. The use of potassium hydroxide instead of sodium hydroxide gives physically softer soaps.

- 6 Name two other substances that manufacturers might add to soaps.
- 7 The sodium and potassium salts of fatty acids. What does this define?
- 8 Tallow is another name for animal fat. True or false?

Soap is basically the sodium (sometimes potassium) salt of a long chain fatty acid. Figure 1 shows two representations of a soap.

1 What is meant by the lower case letter 'n' shown in the top diagram of figure 1?

Soap is a **surfactant** (surface active agent). This is a substance that, when added to a liquid, reduces its surface tension which increases its spreading and wetting properties. Surfactants must be able to deal with both water (polar) and lipid (non-polar) environments.

Soaps have a polar end ($-\text{COONa}$ for the example shown in figure 2) and a non-polar end (a long hydrocarbon chain). The polar end ('head') is **hydrophilic** (literally 'loves water'; *hydro* meaning water, *phile* meaning loving) and it is water-soluble. The non-polar end ('tail') is **lipophilic** and is soluble in lipids. When soap is put into water, the structure breaks up into a positive (Na^+) ion and a negative (carboxylate) ion.

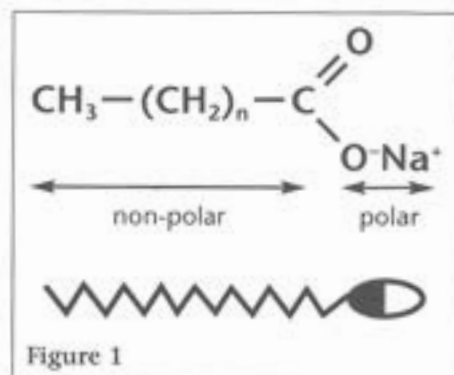


Figure 1

2 What do you think is the literal meaning of *lipophilic*?

3 Sometimes the term *hydrophobic* is used, literally meaning 'fear of water' (*phobos*, meaning fear). Give another term (already used on this page) that could be used to replace the term *hydrophobic* in reference to surfactants.

Surfactants have many uses. They:

- help dyes penetrate fabric evenly when textiles are dyed
- disperse aqueous suspensions of insoluble substances
- help produce aerosols
- aid in froth flotation purification of ores
- act as detergents.

A detergent is something that helps make dirty objects clean (*detergere*, meaning wipe clean). Often the 'dirt' is oil-based and cannot be efficiently removed by water alone. Soaps are detergents; they can dissolve their non-polar ends in lipids and their polar ends in water. The resulting complex is called a micelle (*mica*, grain or crumb). A micelle is a cluster of units, such as globules of grease or oil, surrounded by soap molecules. Detergents reorient their hydrophobic ends away from the water and their hydrophilic ends surround the 'dirt'; thus breaking it down and dissolving it.

4 Figure 2 shows how a soap (or similar detergent) can be used to clean grease from a plate. Refer to the diagrams to help you describe how soap does this. In your answer include the terms: *emulsion*, *hydrophilic*, *lipophilic*, *micelle*, *surfactant*.

5 Explain why figure 2 is not drawn to scale.

6 Give three uses of surfactants, apart from acting as detergents.

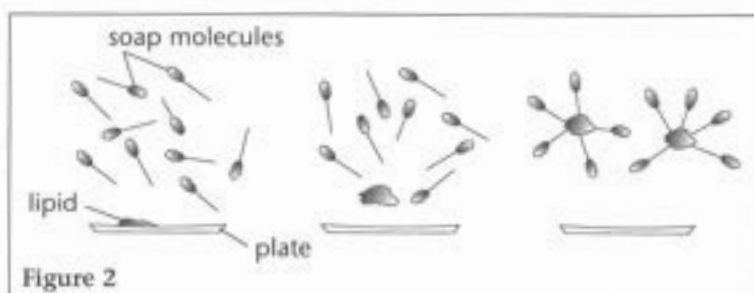


Figure 2

The word *detergent* means a cleaning agent. Synthetic detergents (usually just called detergents) have similar properties to soap but are chemically different. Synthetic detergents do not form a scum with hard water.

7 What is meant by *synthetic*?

8 A substance that increases the wetting and spreading properties of a liquid. What does this define?

9 Define *scum*, in terms of chemistry.

Potential energy is energy that is stored and 'waiting' to be used. Chemical energy, i.e. the energy stored in chemicals, is a type of potential energy. Chemical energy is used to keep living things alive. The energy stored in food is broken down by the body to be used as fuel for the body's metabolic activities. (Metabolism is the sum of all of the chemical processes occurring in the body.) Chemical energy can be converted to heat, electricity and light (and vice-versa). The energy released or absorbed by a chemical reaction can be measured. Chemists tend to focus on measuring changes in heat energy.

- Food energy is broken down to be used to fuel the body's metabolic activities. Name three ways the body uses this energy.
- How can heat changes in a chemical reaction be measured?

Heat energy is sometimes called **enthalpy** (*enthalpy*, meaning to warm or contain heat) and is given the symbol H . (Enthalpy is actually a little more complex than this definition). A change in heat energy is written as ΔH . The symbol Δ (delta) means 'a change'. It is usually not possible to measure the total amount of enthalpy in a system, but it is possible to measure changes. Some reactions give out heat (are *exothermic*), other reactions take in heat from their surroundings (are *endothermic*).

- Figure 1 shows two systems; one is exothermic and the other is endothermic. What do the arrows represent?
- Name three types of energy, apart from heat energy.
- Give an example of a use for each type of energy you named in Q4.

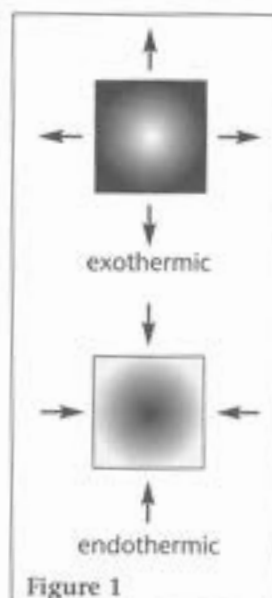


Figure 1

One characteristic of a chemical change is that an energy exchange takes place between the reacting system and its surroundings. The system can either gain or lose energy. Heat energy is measured in joules (J). The convention is that the ΔH (enthalpy change) value for an exothermic reaction is shown as negative (because the system has lost energy) and that ΔH for endothermic reactions is shown as positive (because the system has gained energy).

In a chemical reaction, there are reactants at the start and products at the end. Figure 2 shows the relative energy levels of the reactants and products in two different reactions.

In summary, ΔH – meaning change in enthalpy – is defined as the amount of heat a substance absorbs or emits per unit of that substance (units will vary). The difference between the heat in substances at the start of the reaction should equal the heat in the product plus the heat gained or lost during the reaction.

EXAMPLE

- Begin with 100 kJ; the reaction occurs; end with 85 kJ. Therefore it was an exothermic reaction that gave off 15 kJ of heat energy.

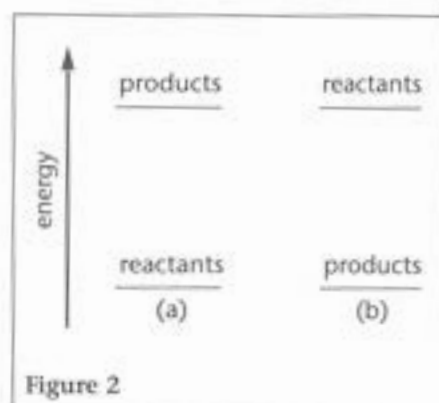


Figure 2

ΔH is equal to the final heat content of the products of a reaction minus the initial heat content of the reactants used.

Endothermic reactions absorb heat (*endo* meaning within), therefore they give a positive value for ΔH . Exothermic reactions emit heat (*exo* meaning out), therefore they give a negative value for ΔH .

- Refer to figure 2 to answer the following.
 - State which reaction is exothermic. Explain your answer.
 - State which reaction is endothermic. Explain your answer.
- An equation describing a reaction stated that $\Delta H = -100$ kJ. What does this tell you about the reaction that occurred?

Calorimetry is a technique used to measure the heat released or absorbed in chemical reactions. A calorimeter is a container (usually insulated) in which a reaction can be carried out and the energy transferred between the system and its environment can be measured. A coffee cup and lid made of Styrofoam™ (expanded polystyrene) make a good calorimeter for measuring some enthalpy changes. (See figure 3.)

- 8 Suggest why a calorimeter is usually insulated.
- 9 Explain why Styrofoam™ is a good material for a calorimeter.

The initial temperature of reactants is measured, the reaction occurs and the maximum change in temperature is noted. The quantity (grams) of reactants is also measured. In an exothermic reaction the temperature will rise and then fall to room temperature.

- 10 Describe the temperature changes in an endothermic reaction.
- 11 Characterised by the evolution of heat. What type of reaction does this describe?
- 12 Mark each statement as true or false.
 - a All reactions involve energy changes.
 - b In a chemical reaction, we begin with reactants that are changed to products.
 - c Calorimeters measure quantities of heat.
 - d Energy can be transferred between systems.
 - e Endothermic reactions give off heat.
 - f Potential energy is chemical energy that has been released.
 - g Living things are fuelled by chemical energy.

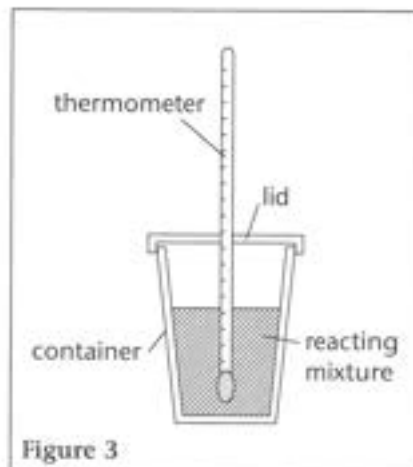


Figure 3

- Research**
- 13 Find out what the *Law of Conservation of Energy* is and write a definition of it.
 - 14 How does measuring ΔH in a reaction illustrate the Law of Conservation of Energy?

When carrying out experiments involving enthalpy changes, it is essential to know the **specific heat capacity** (also called specific heat) of the substances involved. Specific heat capacity can be defined as the amount of heat energy (measured in Joules) required to raise the temperature of one gram of the substance by one degree Celsius (or 1 Kelvin). Water has a specific heat capacity of 4.18. In other words, 1 g of water needs 4.18 J to raise its temperature by 1K or 1°C. The joule is a small unit and kilojoules (kJ) are more likely to be used in calculations. Specific heat can also be defined as the amount of heat energy required to increase one kilogram of a substance.

The enthalpy change of a reaction is calculated using:

$$\text{mass (m)} \times \text{specific heat (C)} \times \text{temperature change } (\Delta T)$$

$$\text{That is, } \Delta H = mC\Delta T$$

- 15 For the equation above, name the quantity that each symbol represents and the appropriate units for that quantity. For example, what does ΔH represent and in what units is it measured?

EXAMPLE

- ii A calorimeter contained 25.0 mL of water at a steady 18.0°C. 5.85 g of sodium chloride was added to the water and the temperature dropped to 15.0°C. Given that the calorimeter constant (heat taken in by calorimeter) is 3.0 J °C⁻¹ and the specific heat capacity of the solution is 4.18 J.g⁻¹.°C⁻¹, calculate the following.
 - a Was heat gained or lost by the system?
 - b What was the actual amount of heat change?
 - c What was the heat of solution (per gram) of sodium chloride under these conditions? (Assume that 1.0 mL of solution has a mass of 1.0 g.)

- a Heat is gained by the system. Heat will flow in from the surroundings to raise its temperature back to room temperature (i.e. it is endothermic, ΔH is positive).

b Write the equation: $\Delta H = mC\Delta T$ and put in the values. (Include the mass of sodium chloride in the solution.)

$$\Delta H = (25.0 + 5.85) \times 3.0 \times 4.18 = 386.9 \text{ J}$$

Account for the heat absorbed by calorimeter, i.e. 3.0 J K^{-1}

$$3 \times 3 = 9 \text{ J},$$

$$\text{total} = (386.9 + 9) = + 395.9 \text{ J}$$

c Alter to heat change per gram:

$$395.9 \text{ J} \div 5.85 = 67.67 \text{ J g}^{-1}$$

16 A student added 12.0 g of a solid to 50.0 mL of water. The room temperature was 21.5°C . The student measured the temperature at regular intervals and noted that the maximum temperature was 24.8°C before the mixture returned to room temperature. Given that the calorimeter constant is $3 \text{ J}^\circ\text{C}^{-1}$ and the specific heat capacity of the solution is $4.18 \text{ J}^\circ\text{C}^{-1} \text{ g}^{-1}$, calculate the following.

- Was heat gained or lost by the system?
- What was the actual amount of heat change?
- What was the heat of solution (per gram) of solute under these conditions?
(Assume that 1 mL of solution has a mass of 1.0 g.)

17 A student added 8.50 g of a solid to 100.0 mL of water. The room temperature was 20.5°C . The student measured the temperature at regular intervals and noted that the minimum temperature was 18.6°C before the mixture returned to room temperature. Given that the calorimeter constant is $3 \text{ J}^\circ\text{C}^{-1}$ and the specific heat capacity of the solution is $4.18 \text{ J}^\circ\text{C}^{-1} \text{ g}^{-1}$, calculate the following.

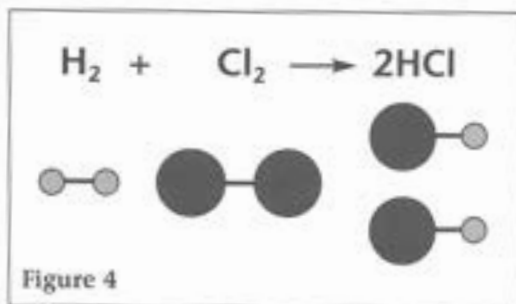
- Was heat gained or lost by the system?
- What was the actual amount of heat change?
- What was the heat of solution (per gram) of solute under these conditions?
(Assume that 1 mL of solution has a mass of 1.0 g.)

18 A firework sparkler contains iron filings in an oxidising mixture. When it is lit, the sparkler throws out glowing specks of iron at 700°C . However, there is less chance of being burnt by these sparks than there is of being scalded by a drop of boiling water at 100°C . Why is this so?

Bond formation in reactions

In chemical reactions, matter is neither created nor destroyed; rather, atoms are rearranged. Before new substances (and new bonds) can be made, bonds have to be broken. It takes energy to break bonds. Energy is released when bonds are made.

Figure 4 shows how the atoms are rearranged when a hydrogen molecule reacts with a chlorine molecule to form two hydrogen chloride molecules.



19 Use figure 4 to help you describe how bonds are broken and made in the synthesis of hydrogen chloride.

20 The synthesis of hydrogen chloride is an exothermic reaction. In this reaction, what is the relationship between the energy required to break bonds and the energy released in making bonds?

21 Write the correct term for each definition.

- A technique used to measure the heat gained or lost in a chemical reaction: _____
- The sum total of the chemical reactions proceeding in an organism: _____
- A container in which heat experiments are carried out: _____
- Can be very simply expressed as the ratio of heat absorbed by one gram of a substance to the temperature change caused by this heat: _____

Potential energy diagrams are very useful in studying energy changes of reactions. They show:

- energy levels of reactants
- energy levels of products
- activation energy
- enthalpy change for the reaction.

Energy must be added to start a reaction. This is because an **energy barrier** (e.g. to break bonds) must be overcome. An energy barrier separates substances used in a reaction (reactants) from the products created. Overcoming the barrier to enable one substance to be converted into another requires the input of energy. This energy input is called the **Activation Energy (E_a)**. Activation energy is the minimum amount of energy required to allow a reaction to occur; for example, the head of a match will burst into flame when it is struck – the heat from the friction starts the reaction.

- 1 Name another reaction that requires energy to start it.

For some reactions the energy available at room temperature is enough to overcome the activation energy. These reactions are said to be **spontaneous**. Iron rusting is a spontaneous reaction.

- 2 Define *spontaneous*.
- 3 Describe one other spontaneous reaction.

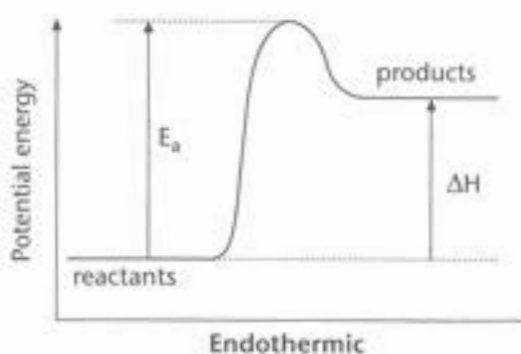
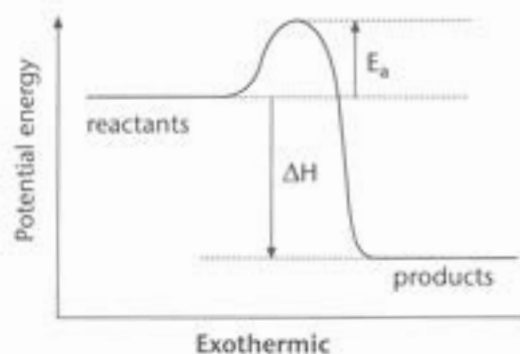


Figure 1

Figure 1 has two potential energy diagrams; one showing an exothermic reaction and the other showing an endothermic reaction.

- 4 Use figure 1 to help you answer the following.
 - a In the exothermic reaction, which is at the higher energy level: products or reactants?
 - b In the exothermic reaction, has energy been taken in or released by the reaction?
 - c In the exothermic reaction, is ΔH positive or negative?
 - d In the endothermic reaction, which is at the higher energy level: products or reactants?
 - e In the endothermic reaction, has energy been taken in or released by the reaction?
 - f In the endothermic reaction, is ΔH positive or negative?
- 5 Which is more likely to be a spontaneous reaction at room temperature: a reaction with a high activation energy or a reaction with a low activation energy? Explain your answer.

Standard conditions

All values quoted in references such as textbooks are usually referring to the reaction carried out under standard conditions. Two specific terms are **Standard Laboratory Conditions (SLC)** and **Standard Temperature and Pressure (STP)**. Common temperatures used as standards are 0°C (273K) and 25°C (298K) and generally pressures are around 101.3 kPa (1 atmosphere).

- 6 How do you convert degrees Celsius ($^\circ\text{C}$) to Kelvins (K)?
- 7 The minimum amount of energy required to convert a stable system into a reactive system. What does this define?

- 8 Mark each statement as true or false.
- a Activation energy is energy given out in a reaction.
 - b In chemical reactions, only bond-making takes place.
 - c ΔH is the shorthand way of representing enthalpy change.
 - d Energy is required to start all chemical reactions.
 - e The temperature 273 Celsius ($^{\circ}\text{C}$) is the same as 0 Kelvin (K)

Catalysts

Catalysts are often used in industry. A catalyst acts by altering the path of the reaction and changing the activation energy required. Note that activation energy is **always** positive. Figure 2 shows the effect of a catalyst on a reaction. The solid lines represent the uncatalysed reaction and the broken lines represent the catalysed reaction.

A catalyst alters the rate of reaction (usually speeding it up) or allows it to proceed at a lower temperature. Catalysts are not used up during a reaction and are unchanged at the end of the reaction; therefore, they are 'recycled'.

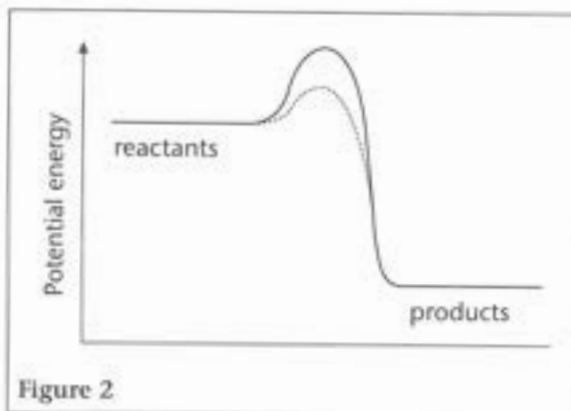
- 9 Copy figure 2. On the graph, draw and label each of the following.

- a The activation energy for the uncatalysed reaction
- b The activation energy for the catalysed reaction
- c The enthalpy change (ΔH) for the reaction.

- 10 Is the reaction shown in figure 2 endothermic or exothermic? Explain your answer.

- 11 A student added manganese dioxide (a black insoluble solid) to hydrogen peroxide (a clear colourless liquid). Oxygen was given off and there seemed to be some black solid remaining. Describe how the student could test whether the manganese dioxide catalysed the reaction or whether it was a reactant itself.

- 12 What are the standard conditions under which heats of reactions are usually quoted?



Catalysts can also allow reactions to take place under milder (e.g. less heat and pressure) conditions, which saves energy. This doesn't give more products than the uncatalysed reaction does – just the same amount, but more quickly. Many transition metals and their compounds are catalysts.

Enzymes

Enzymes can be thought of as **biological catalysts**. They are much more specific than inorganic catalysts and only work at specific temperatures and pH levels. Enzymes act by forming temporary bonds with the molecules that are involved in a chemical reaction. They are usually made of complex proteins and can be **denatured** (their properties destroyed) at high temperatures. Enzyme names often end in '-ase' and refer to the reaction they catalyse. For example, lipases are enzymes that split up lipids (fats and oils). Life as we know it would not exist without enzymes. Enzymes allow metabolic processes to react many times faster than would normally be possible. Enzyme reactions also occur at much lower temperatures than would otherwise be the case. This is important in cells because without enzymes the heat energy required for many reactions would be too high for the cell to survive. Enzymes are also used in processes such as making bread, beer and cheese.

- 13 Outline how catalysts work to speed up reactions.
- 14 Describe a similarity between enzymes and inorganic catalysts.
- 15 Describe a difference between enzymes and inorganic catalysts.
- 16 Give two reasons why a manufacturer might use catalysts in industrial reactions.

Generally changes of state are **reversible**; for example, liquid water can be changed to solid ice and vice versa. Most chemical reactions are theoretically reversible. This means that not only do reactants form products; but (depending on conditions) products can be changed back to the original reactants. Equilibrium reactions are special types of reversible reactions.

Alexander William Williamson (1824–1904) was the first person to systematically study reversible reactions. He was also the first person to come up with the concept of **dynamic equilibrium**. Dynamic equilibrium describes the equilibrium, or balance, of forward and reverse reactions happening simultaneously. Therefore an equilibrium reaction involves two reactions that are occurring at the same time but in opposite directions. In other words, reactants are being changed to products at the same time that products are being changed to reactants. At equilibrium, the rates of the forward reactions and reverse reactions are the same. This is shown by using the equilibrium symbol ($\rightleftharpoons$).

For example: Products $\rightleftharpoons$ Reactants

- 1 Write in words the equation for each reaction.
 - a $\text{CaCO}_3(\text{s}) \xrightarrow{\text{heat, open system}} \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 - b $\text{CaO}(\text{s}) + \text{CO}_2(\text{g}) \xrightarrow{\text{pressure, closed system}} \text{CaCO}_3(\text{s})$

Some reactions can be reversed by changing reaction conditions. For example, if calcium carbonate is heated in a closed vessel, at first the only reaction will be calcium carbonate decomposing to form calcium oxide and carbon dioxide. When the pressure and the concentration of calcium oxide and carbon dioxide reach a certain point, they start to react to form calcium carbonate. This is shown by the equation: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

- 2 Will an equilibrium reaction in a sealed system ever reach completion, i.e. give 100% products? Explain your answer.

Summary

- Reactants combine to make products. (The forward reaction.)
- In some reactions, products can break apart and turn back into the reactants. (The reverse reaction.)
- At equilibrium, forward and reverse reactions are taking place at the same rate.
- At equilibrium it *seems* as if no reactions are occurring.
- Equilibrium reactions do not go to completion; at equilibrium there is a mixture of reactants and products.

When talking about equilibrium reactions, the terms **micro** and **macro** are often used. At equilibrium, there are reactions taking place that are only visible at the microscopic level. At the larger (macro) level, the naked eye can't see any change.

- 3 Suggest why it seems that no reactions are occurring at equilibrium.
- 4 Define *micro* and *macro*.
- 5 Study the following description of an equilibrium reaction and sketch a graph to show how this reaction proceeds. Graph concentration on the y-axis and time on the x-axis.

In an equilibrium reaction, at the moment the reactants are mixed the concentration of reactants is 100% and the concentration of products is 0%. As time passes, the concentration of reactants decreases, reaches a minimum point and stays at that concentration. At the same time the concentration of products increases, reaches a maximum point and stays at that concentration.

- 6 Copy this statement and include the missing word:
For a chemical reaction at _____, the amount of reactants being transformed to products per unit time is exactly matched by the amount being transformed to reactants (from products) per unit time.
- 7 The nature and properties of the equilibrium state remain the same, regardless of the direction from which it is reached. Explain what is meant by this statement.

Catalysts and equilibrium reactions

Figure 1 shows the reaction profile for an exothermic equilibrium reaction. You can see that the activation energy (E_a) for the catalysed reaction is less than the activation energy for the uncatalysed reaction. Catalysts speed up both the forward and reverse reactions equally. This means that adding a catalyst to an equilibrium reaction causes that reaction to reach equilibrium more quickly, but does not alter anything else.

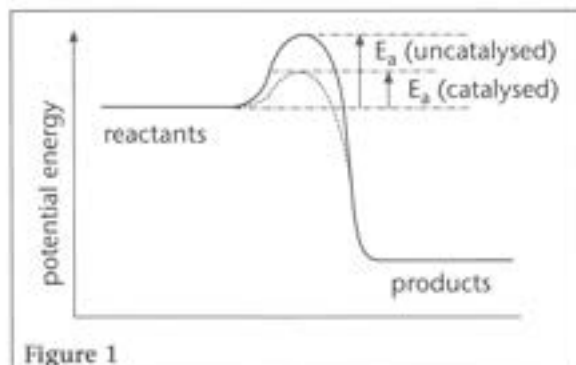


Figure 1

Changing equilibrium systems

The position of equilibrium varies from reaction to reaction. Often chemists try to affect equilibrium reactions. Reasons for this include speeding up the reaction and increasing the yield of products.

8 Name three things a chemist could do to affect an equilibrium reaction.

Henri-Louis Le Châtelier (1850–1936) made great contributions to our understanding of equilibrium systems. He is famous for **Le Châtelier's principle**. One version of his principle is: *A system in equilibrium, when subjected to a stress resulting from a change in temperature, pressure or concentration, and causing the equilibrium to be upset, will adjust its position of equilibrium to relieve the stress and re-establish equilibrium.*

This can be summarised as follows.

- An equilibrium system will not change its rate of forward and reverse reactions until conditions are changed.
- Equilibrium reactions can be affected by changes in concentration, heat and (when gases are involved) pressure.
- An equilibrium reaction will change in order to reduce the stress on the system.

It doesn't matter how the conditions of the system change, there are only two possible outcomes:

- The forward reaction speeds up to produce more products
- The reverse reaction speeds up to produce more reactants.

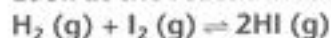
So when conditions change, the forward/reverse reactions change; after some time, the reaction will reach a **new equilibrium point**.

Changes in concentration

If the concentration changes in the product or the reactant, the forward and reverse reactions will alter to minimise this change.

EXAMPLE

i Look at the reaction between hydrogen and iodine forming hydrogen iodide:



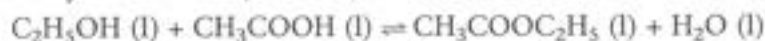
If hydrogen is added to the above reaction at equilibrium point, the forward reaction will increase in speed (in order to remove 'excess' hydrogen and produce more hydrogen iodide) until a new equilibrium is reached.

If hydrogen iodide is added at equilibrium point, the reverse reaction will speed up (in order to remove 'excess' hydrogen iodide and produce more hydrogen and iodine) until a new equilibrium is reached.

9 Use Le Châtelier's principle to predict what would happen to the above equilibrium system if hydrogen iodide was removed.

10 What would happen to the reaction between hydrogen and iodine forming hydrogen iodide if a catalyst was used? Explain your answer.

Esterification is an equilibrium process. One example is when ethanol reacts with ethanoic (acetic) acid to form ethyl ethanoate (an ester) and water.



11 Outline two ways that the yield of ester could be increased.

Changes in heat

Reactions that give out heat are exothermic (i.e. ΔH is negative). Reactions that take in heat are endothermic (i.e. ΔH is positive). Heating or cooling equilibrium reactions will have different effects, depending on whether these reactions are exothermic or endothermic. Consider heat as a reactant or product in reactions (not strictly true, but a useful simplification). This can help in using Le Châtelier's principle to predict changes due to heat. A reaction that is exothermic going from reactants to products (forward reaction) will be endothermic going from products to reactants (reverse reaction). If heat is a 'product' then adding more heat will cause the reaction to try to remove this heat.

- Adding heat to an exothermic reaction (that 'wants' to lose heat) will oppose the forward reaction and speed up the reverse (endothermic) reaction.
- Adding heat to an endothermic reaction (that 'wants' to gain heat) will speed up the forward reaction and oppose the reverse (exothermic) reaction.

12 Copy each reaction and use Le Châtelier's principle to predict what would happen if the temperatures were changed from standard conditions.

- | | | | |
|---|---|-------------------------------|----------|
| a | $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ | $\Delta H = -58.1 \text{ kJ}$ | (heated) |
| b | $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g})$ | $\Delta H = +131 \text{ kJ}$ | (heated) |
| c | $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ | $\Delta H = -197 \text{ kJ}$ | (cooled) |
| d | $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ | $\Delta H = +182 \text{ kJ}$ | (cooled) |

Changes in volume

Changes in pressure can influence equilibrium reactions where gases are present. However, there also needs to be a difference in the number of moles of products compared to reactants. If pressure is changed, Le Châtelier's principle says that the system will react to try to restore original pressure. Increasing pressure causes a decrease in the volume of a gas; decreasing pressure causes an increase in the volume of a gas (assuming temperature remains constant).

- An increase in pressure will favour the reaction that decreases volume.
- A decrease in pressure will favour the reaction that increases volume.
- Solids and liquids present are assumed to take up minimal volume.

EXAMPLE

ii Study this reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

The reactants take up 3 volumes (of a given unit) and the products take up 2 volumes. In other words a volume ratio of 3:2.

Therefore an increase in pressure would favour a decrease in volume, driving the equilibrium system from left to right.

13 Copy each reaction and use Le Châtelier's principle to predict what would happen if the pressure was changed from standard conditions.

- | | | |
|---|---|----------------------|
| a | $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ | (pressure increased) |
| b | $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g})$ | (pressure increased) |
| c | $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ | (pressure decreased) |
| d | $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ | (pressure decreased) |

14 Summarise Le Châtelier's principle.

15 What is a prediction?

Research **16** In science you will hear about *Theories* and *Principals* or *Laws*. Theories are different from Principals/Laws. Find out how they are different.

In this assignment, you must show that you have used a variety of sources. Include information from textbooks, periodicals, CD ROMs, the Internet and encyclopaedias.

Don't forget to include a bibliography!

- 1 The scientists listed below all worked in chemistry. Describe briefly (five or six sentences each) the major contributions of each scientist. Include where and when their research was carried out.

a James Chadwick	b Neils Böhr
c John Dalton	d JJ Thomson
e Ernest Rutherford	f Dmitri Menedeléeu (Note that spellings vary.)
g Amadeo Avogadro	h Gay Lussac
i Democritus	j Robert Bunsen
- 2 Name one person involved in chemistry who was not mentioned in the previous question. Outline his or her contribution to chemistry.
- 3 Give a brief definition (two or three lines) for each term.

a ion	b sub-atomic particle
c molecule	d isotope
e Periodic Table	f polymer
g organic	h acid
i base	j atom
k alcohol/alkanol	l pH
m octet	n mole
- 4 Chemistry is one of the categories for the Nobel Prize. Find out:

a what the Nobel Prize is
b how it began
c why it is awarded
d the name of a scientist or scientific team who was awarded the Nobel Prize in chemistry and the year the prize was awarded.
- 5 The Kelvin is a scale of temperature. Define the Kelvin in chemistry terms and give a brief description of the scientist after whom it was named.
- 6 A law is a statement that is considered to be true. There are basic laws in chemistry. These include the *law of conservation of mass* and the *law of definite proportions* (sometimes called the *law of constant composition*). Give brief definitions (one or two lines) for these two laws.
- 7 Match each term to the correct definition.

a titration	i means two (containing two elements)
b polar	ii the ability of a system to do work
c energy	iii attraction between oppositely charged particles
d binary	iv process to determine concentration of a solution
- 8 Chemists are employed in many areas of society, such as education, industry, medicine and government institutions. Research one area in which chemists contribute to our way of life. In your answer, state the precise area of chemistry and describe what chemists in this particular occupation do.
- 9 Explain the difference between a chemist and a pharmacist.

Write the element each symbol represents to complete the crosswords.

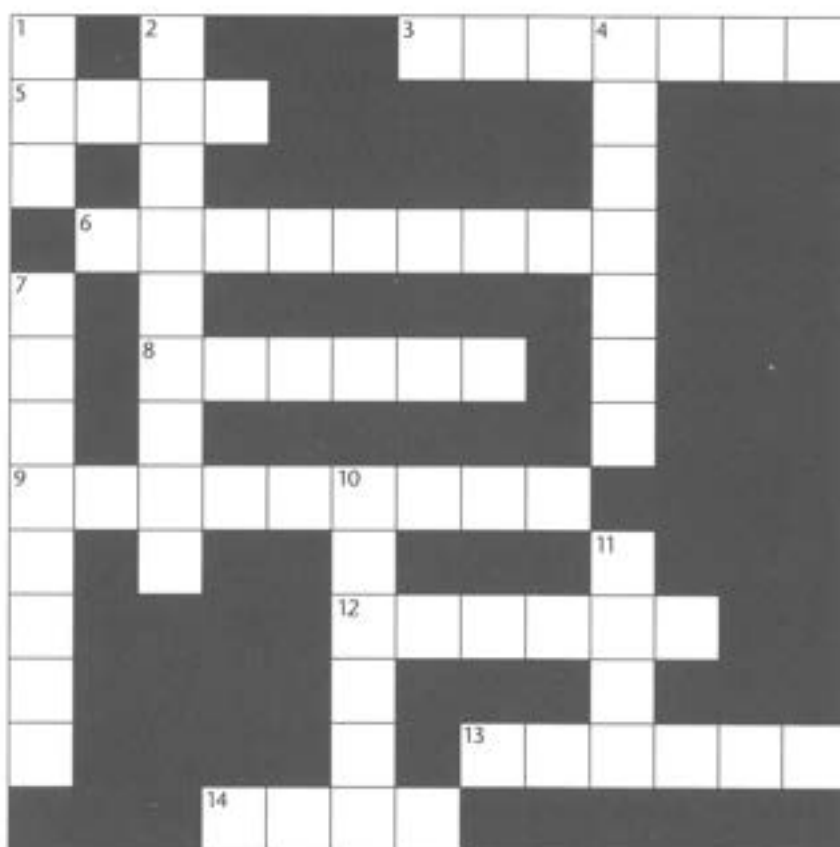
Metals

Across

- 3 Ca
5 Fe
6 Mg
8 Ag
9 Al
12 Cu
13 Na
14 Au

Down

- 1 Sn
2 K
4 Cd
7 V
10 Ni
11 Pb



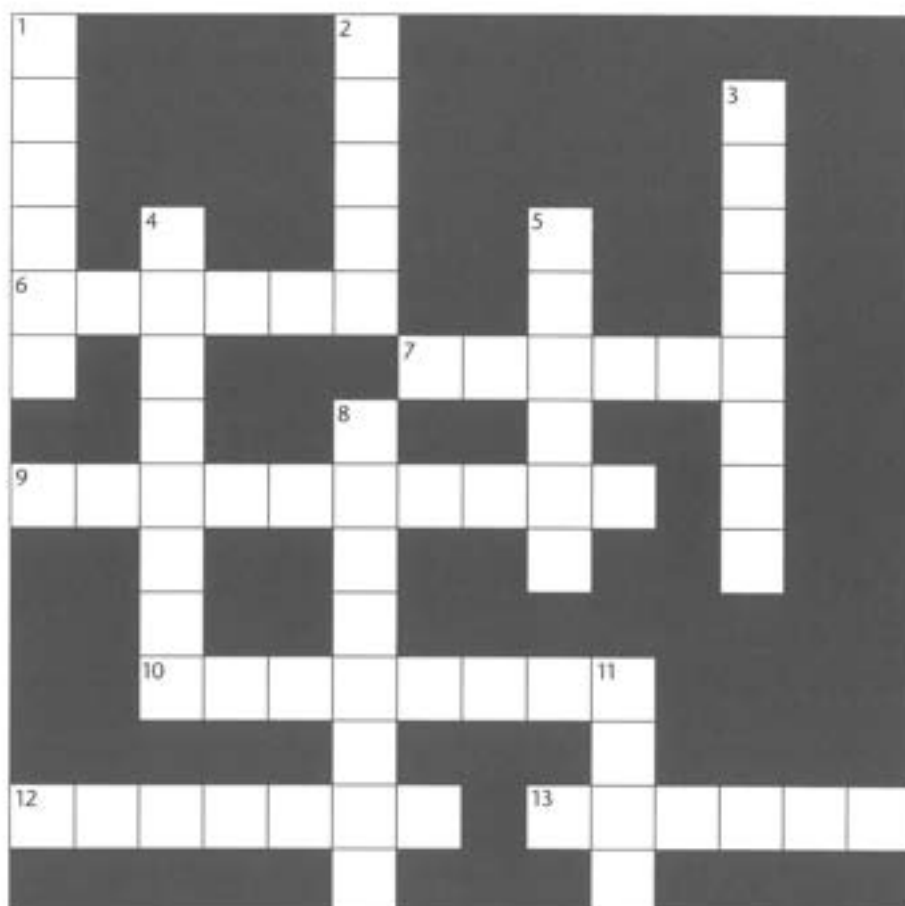
Non-metals

Across

- 6 O
7 S
9 P
10 N
12 Br
13 I

Down

- 1 C
2 Ar
3 F
4 H
5 He
8 Cl
11 Ne



SELECTED ANSWERS 1–10

1 The mole

(Note that individual numeric answers may vary slightly, depending on which Periodic Table or Data Book students refer to for atomic masses/weights.)

- 1 e.g. Hat trick (cricket), square (house areas), bale (of wool) 2 a 14.6×10^{23} b 1.99×10^{-15} 3 The particular atom of carbon that has 6 protons and 6 neutrons
4 a O, 16.00 b Ne, 20.18 c Na, 22.99 d Cu, 63.55
5 a 32.00 g b 28.02 g c 70.90 g 6 a 44.01 g
b 208.2 g c 102.0 g d 180.2 g e 98.09 g f 74.10 g
g 60.05 g 7 a CO_2 , 22.01 g b HCl, 3.646 g c H_2SO_4 , 196.18 g d PCl_5 , 208.2 g e $\text{Mg}(\text{NO}_3)_2$, 148.3 g
f NH_3 , 17.03 g g Cl_2 , 70.90 g 8 a 2.998 b 0.2503
c 1.001 d 0.5000 e 1.998 f 0.1003 g 0.0999
9 $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ 10 a CH_2O b C_2H_5
c CH_2O d Al_2O_3 11 a CH_2 b C_8H_{16} 12 N_2O_3 and
 N_2O_5 13 a 0.32 g b MgO 14 $\text{C}_6\text{H}_{12}\text{O}_6$ 15 a CH_3O
b $\text{C}_2\text{H}_6\text{O}_2$ 16 $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ 17 a True b False
c True d True e False

2 Moles and gases

- 1 0.9099 moles 2 8.000 g 3 0.0786 moles 4 1.080 g
5 1156 L 6 a $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
b 2 moles c 100 L 7 a $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{SO}_3(\text{g})$ or
 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ b 0.5 moles c 25 L

3 Concentrations of solutions

- 1 a 500 g L^{-1} b 10 g L^{-1} c 100 g L^{-1} d 37.5 g L^{-1}
2 6×10^{-1} 3 292.2 g 4 24.00 g 5 a 0.1000 M
b 0.0999 M c 0.1 M 6 a 24.52 g b 12.26 g
c 73.56 g 7 Mass/mass describes the mass of solute
dissolved in a particular mass of solvent. Mass/volume
relates to the mass of solute dissolved in a given volume
of solvent. 8 Number of moles of solute dissolved per
litre of solution

4 Dissolving and dissociating

- 1 Each particle is bonded to many others in a three-dimensional formation. 2 More energy must be given out by making bonds than taken in by breaking bonds.
3 a Ionic b Polar – it dissolves sodium chloride, which is ionic. 4 a $\text{KF}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{K}^+(\text{aq}) + \text{F}^-(\text{aq})$
b $\text{MgCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$
c $\text{K}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{K}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
5 Very strong bonds between ions 6 No ionisation/dissociation 7 Hydrochloric acid
8 $\text{HNO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
9 Molecule too large, bonding inside molecule too strong

5 Solutions, suspensions

- 1 Dissolved 2 To do with water 3 The same composition throughout 4 If no individual particles can be seen it must have dissolved. 5 No more solute can be dissolved. 6 Immiscible 7 e.g. Paint, muddy water
8 Particles gradually fall to the bottom of the liquid.
9 e.g. Hair spray, deodorant, paint, shaving foam
10 Both have particles dispersed in air, smoke is solid particles; fog is water droplets in air 11 e.g. Gelatine,

fruit jelly, hair gel, some toothpastes 12 Surrounding substance 13 e.g. Smoke, natural rubber 14 a Yes, fog is a suspension. b No, sugar dissolved in water is a solution. 15 Cooled rapidly to below melting point so crystals aren't formed 16 Movement of small particles in a medium being bumped/jostled in a random manner by the particles of the medium

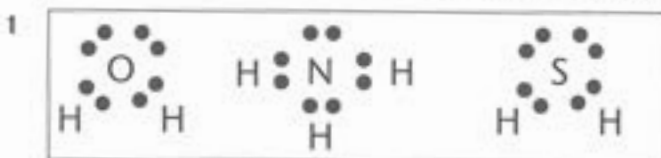
6 Handling liquids

- 1 Beaker, J 2 Burette, B 3 Test tube, H 4 Measuring cylinder, I 5 Conical flask, D 6 Dropper, F 7 Pipette, C 8 Funnel, E 9 Volumetric flask, G 10 Stirring rod, A

7 Solubility and precipitation

- 1 a Copper sulfate, soluble b Calcium carbonate, insoluble c Silver chloride, insoluble d Ammonium hydroxide, soluble e Lead (IV) nitrate, soluble
f Lithium phosphate, insoluble 2 a No precipitate
b $\text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{PbCl}_2(\text{s})$ lead (II) chloride
c $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$,
 $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ copper (II) oxide
3 Filtration 4 a Silver chloride, AgCl, 0.005 moles
b Iron (II) hydroxide, $\text{Fe}(\text{OH})_2$, 0.1 moles 5 a Magnesium carbonate, MgCO_3 , 0.02 moles b Barium sulfate, BaSO_4 , 0.016 moles 6 a Ion with a negative charge
b Ion with a positive charge c Insoluble substance formed when two solutions are mixed

8 Polar nature of water and similar molecules



- 2 Hydrogen 3 Oxygen 4 Slight separation of charge; two separate charged areas 5 Should show δ^+ and δ^- aligned 6 Teacher to check; axes should be labelled, points plotted in pencil, etc. 7 Ammonia, water and hydrogen fluoride 8 a NH_3 b H_2S 9 Outer or bonding electrons

9 Electronegativity

- 1 e.g. The extent to which an atom can attract a share of the electron pair forming a covalent bond 2 Donate (to non-metal) to form ionic bond 3 a Noble gases, full outer shell b Teacher to check c Increase d Decrease
4 a Ionic b Covalent c Polar covalent 5 a True
b False c True

10 Periodic properties 1

- 1 One (positive) proton more than electrons
2 a Helium, He b Neon, Ne c Argon, Ar 3 Noble gases, Group VIII (0) 4 a Lithium, Li b Sodium, Na
c Potassium, K 5 Group I, alkali metals 6 a Fluorine, F
b Chlorine, Cl 7 Group VII, halogens 8 Helium
9 Potassium 10 Increase 11 Decrease 12 Forms positive ions readily 13 a True b False c True
d False e True f False g True h True

SELECTED ANSWERS 11–17

11 Periodic properties 2

1 Any two from noble gases (Group VIII), lanthanides, actinides 2 a Decreases b Increases 3 Extra electron shells 4 Teacher to check 5 a Decrease then increase b Decrease c Increase then decrease 6 e.g. Thermal and electrical conductivity, lustre 7 Silicon, also in Group IV 8 Noble gases 9 Increasing numbers of protons causes increasing attraction for electrons in outer shell 10 Helium 11 Carbon

12 Dependability of information

1 Two from Internet, magazines, CD ROM, other people 2 Someone may have already carried out the investigation 3 a e.g. Agronomist, fertiliser company, relevant textbook b e.g. Chemist, textbook 4 More personal, possibility of bias/inaccuracy 5 b e.g. Temperature, size of container 6 Hypothesis: Increasing concentration increases the rate of reaction; Independent variable: concentration; Dependent variable: time for gas to be given off; Controlled variables: mass of zinc, volume of acid 7 Conforms to a standard 8 Teacher to check 9 A stopped clock is accurate twice a day; a late clock is never accurate 10 a Slight inaccuracy is acceptable in day to day measurements, e.g. 'It takes 10 or 12 minutes to walk there.' b 100% accuracy is needed when less than that can cause problems; e.g. locating the site of a small buried treasure from longitude and latitude coordinates. 11 Can be depended upon 12 Teacher to check 13 The bus is always 5 minutes late. 14 e.g. How justified something is; to be sound and support the claim 15 Teacher to check 16 They questioned whether his passport was acceptable or real. 17 There are often disputes between scientists and they often reach different conclusions based on the same findings. Also, doubt could arise about factors such as the methodology of experiments. 18 a True b True c False d True

13 Metals

1 Conducts heat efficiently 2 They are all unreactive. 3 a Shiny b Can be shaped c Can be drawn into a wire d Snaps easily e Mineral from which substances such as metals can be obtained profitably f A rock of definite chemical composition 4 a Iron (III) oxide b Hydrated aluminium oxide c Lead sulfide d Calcium carbonate e Hydrated, basic copper carbonate f Calcium oxide 5 a Iron, Fe and oxygen, O b Aluminium, Al and oxygen, O c Lead, Pb and sulfur, S d Copper, Cu, carbon, C and oxygen, O e Hydrogen, H and oxygen, O f Nitrogen, N and hydrogen, H 6 Once they are used up, no more will be formed 7 e.g. To dig into the ground and remove materials 8 e.g. Creates pollution, displaces wildlife, disrupts environmentally sensitive areas 9 e.g. Coal, iron ore, aluminium ore, gold 10 e.g. Creates jobs, provides goods or the raw materials for manufacturing goods 11 In a rough/impure state, needs processing to obtain the required substance

14 Extraction and refining of metals

1 a Ability to cause change or to do work b Compound containing sulfur c To do with floating d Chemical used in a reaction e Undergoes chemical reactions readily 2 Teacher to check 3 They can weaken or cause unfavourable reactions in the product being produced. 4 Teacher to check 5 a $\text{Cu}_2\text{S}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{Cu}(\text{s}) + \text{SO}_2(\text{g})$ b $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 2\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g})$ c $\text{NaCl}(\text{s}) \rightarrow \text{Na}(\text{s}) + \frac{1}{2}\text{Cl}_2(\text{g})$ or $2\text{NaCl}(\text{s}) \rightarrow 2\text{Na}(\text{s}) + \text{Cl}_2(\text{g})$ d $\text{Al}_2\text{O}_3(\text{s}) \rightarrow 2\text{Al}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g})$ or $2\text{Al}_2\text{O}_3(\text{s}) \rightarrow 4\text{Al}(\text{s}) + 3\text{O}_2(\text{g})$ 6 Metal mixed (often initially in molten form) with other substance(s) – usually other metals 7 Copper and tin easier to obtain from ores than iron (Also some copper and tin occurred as native metals in nature.) 8 a Smelt b Ingot c Slag d Alloy e Electrolysis f Haematite g Furnace h Reduction i Ore

15 Properties and uses of metals

1 a $\text{Cu}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CuO}(\text{s})$ or $2\text{Cu}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CuO}(\text{s})$ b $2\text{Fe}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g}) \rightarrow \text{Fe}_2\text{O}_3(\text{s})$ or $4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$ 2 a $\text{Na}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NaOH}(\text{aq}) + \frac{1}{2}\text{H}_2(\text{g})$ or $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$ b $\text{K}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{KOH}(\text{aq}) + \frac{1}{2}\text{H}_2(\text{g})$ or $2\text{K}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{KOH}(\text{aq}) + \text{H}_2(\text{g})$ c $\text{Ca}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{aq}) + \text{H}_2(\text{g})$ 3 a $\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$ b $\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$ c $\text{Mg}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2(\text{g})$ 4 e.g. Collect, melt, remove impurities, cast into ingots. Should cost a fraction of extracting from ore 5 e.g. Less damage to environment, less mining 6 Light, strong, easily shaped 7 Improved properties 8 Teacher to check

16 Metals: terms and research

1 n 2 d 3 q 4 j 5 o 6 l 7 t 8 e 9 b 10 f 11 x 12 a 13 w 14 u 15 v 16 s 17 c 18 m 19 r 20 p 21 k 22 g 23 i 24 h 25 Teacher to check

17 Local water supplies

1 Teacher to check 2 Local council, water company/authority 3 Major source of water supplies 4 Includes fertiliser, pesticides, industrial waste, etc. 5 a e.g. Atomic absorption spectroscopy b Teacher to check, e.g. precipitation, flame tests c Indicators, pH meter d e.g. Amount of soap to lather, dissolved solids 6 e.g. Depends how toxic they are 7 How water is carried (via pipes) to houses and industry 8 e.g. Particles get caught between grains 9 Teacher to check 10 Potassium alum sulfate dodecahydrate 11 O_3 12 Less toxic, breaks down to oxygen 13 To reduce tooth decay 14 Calcium oxide 15 Sodium

SELECTED ANSWERS 17–25

carbonate 16 Membrane – thin porous sheet; Eliminate – get rid of; Pollutants – harmful impurities; Polymers – large molecules made up of similar repeating units; Micron – a millionth of a metre; Pathogen – something that causes a disease

18 Hard water

1 a Measuring cylinder/pipette b Burette 2 e.g. Shake for the same length of time, same amount of lather, same quantities 3 Teacher to check 4 Sodium carbonate is more effective using smaller quantities; Softa eventually removes more hardness. 5 Use a soapless detergent. 6 Large amounts of soap needed, hard deposits in hot water systems and kettles 7 Temporary can be removed by boiling, permanent can't 8 Precipitation 9 Too expensive, too much energy required to boil water 10 Salt solution 11 Water with less than 50 mg L⁻¹ of total hardness is regarded as soft and over 200 mg L⁻¹ of hardness is regarded as hard. 12 Taste, essential minerals

19 Water quality

1 Teacher to check; include the smaller the depth, the greater the turbidity 2 The smaller the amount of light transmitted, the greater the turbidity. 3 Figure 2 is more accurate, uses specific measurements rather than judgements with eye. 4 Burette or pipette 5 0.25 g, 0.33 g, 0.80 g, 3.35 g, 0.03 g 6 No water left, no solid lost, no decomposition of residue 7 2500, 3300, 8000, 33500, 300 8 Teacher to check 9 a 8 ppm (approx.) b 100 ppm (approx.) c 210 ppm (approx.) d 670 ppm (approx.) 10 KCl 11 Record and store readings 12 To manipulate and display results 13 Too much dissolved solid 14 Linear relationship between conductivity and dissolved solids 15 Convenient, not based on any intrinsic scale 16 To give a scale 17 Cloudiness, lack of clarity 18 Much quicker, but need to set up range of standards 19 e.g. An instrument for measuring cloudiness in a liquid. 20 Will vary, but generally accepted that it should be less than 2–3 g per litre. 21 To kill germs; in too low a concentration

20 Water quality and oxygen

1 Photosynthesis 2 At surface near e.g. waterfalls 3 Teacher to check 4 As temperature increases, solubility decreases 5 3.45 g 6 e.g. Find the concentration of a substance by reacting it with a known amount of another reagent 7 Teacher to check; include equipment, taken at same place, etc. 8 One that exists without oxygen 9 Further away from surface and atmosphere 10 Poisonous 11 Process where electricity breaks down a liquid

21 Biological oxygen demand

1 400 mg L⁻¹ 2 Suffocate 3 Biochemical oxygen demand 4 As time passes organics are oxidised to reach a minimum, but not all organics are oxidised. 5 The lower the BOD, the better 6 Biodegradable 7 Maintain at a steady temperature 8 Teacher to check

22 Viscosity

1 The curve on the surface of liquid 2 The tap is opened until the required amount of liquid is released. 3 Water, 45; milk, 19; ethanol, 55; cooking oil, 122 4 Water, 25 sec; milk, 32 sec; ethanol, 29 sec; cooking oil, 76 sec 5 Water, ethanol, milk, cooking oil 6 Avoid contamination 7 Stopwatch, retort stand/boss head/clamp (to hold burette steady) 8 a Increases b Increases 9 Time how long it takes for an object to fall through the liquid; the longer it takes, the greater the viscosity. Drop the same object from the same height, have the same container, same temperature, etc. 10 e.g. Honey, syrup, lubricating oil 11 a True b False c True d True e True f False g True h True

23 Saponification

1 Fat is solid and oil is liquid at room temperature 2 Equilibrium reaction, doesn't go to completion 3 1, 2, 3-propanetriol 4 Teacher to check; the R group should have 17 carbons 5 Teacher to check; the R group should have 15 carbons 6 Perfume, colour, moisturiser, etc. 7 Soaps 8 True

24 Properties of soap

1 A large number 2 'Likes' fats and oils 3 Lipophilic 4 Teacher to check; should include reference to the diagram and the use of terms mentioned 5 Detergent molecules are too small 6 e.g. To disperse weedkillers, dyes and perfumes in aqueous solutions to make them disperse more easily, helping clean up oil spills. 7 Made by humans/artificial 8 Surfactant 9 Froth, foam or crust formed over a substance

25 Energy and reactions

1 e.g. Growth, repair, daily activities 2 Using a thermometer to measure temperature changes 3 Direction of heat out of or into the systems 4 e.g. Chemical, electrical, potential, mechanical/kinetic, light 5 Teacher to check 6 a (b) Products are at a lower energy level than reactants b (a) Products are at higher level than reactants 7 It was exothermic and 100 kJ of energy was released 8 To minimise heat exchanged with surroundings 9 Good insulator 10 Decreases and then rises to room temperature 11 Exothermic 12 a True b True c True d True e False f False g True 13 Energy is neither created or destroyed but remains constant in a system. 14 Energy in reactants equals energy in products plus whatever is given off or absorbed. 15 ΔH = enthalpy change (J), m = mass (g), C = specific heat (J.g⁻¹), ΔT = temperature change (°C or K). If kg is used for mass then ΔH is in kJ. 16 a Lost b 865 J c 72.2 J.g⁻¹ 17 a Gained b 690 J c 81.1 J.g⁻¹ 18 The mass of the speck is very small so the total heat is also minute. 19 1 hydrogen – hydrogen bond broken; 1 chlorine – chlorine bond broken; 2 hydrogen – chlorine bonds formed 20 More energy released making bonds than put in to break bonds 21 a Calorimetry b Metabolism c Calorimeter d Specific heat

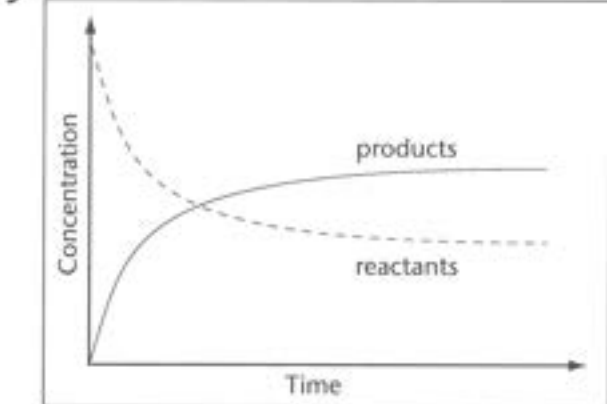
SELECTED ANSWERS 26–29

26 Reaction profiles

1 Lighting a Bunsen burner 2 Happens without outside interference 3 Any that occurs once reactants are mixed, e.g. acid plus active metal 4 a Reactants b Released c Negative d Products e Taken in f Positive 5 Low; less energy needed 6 Add on 273. 7 Activation energy 8 a False b False c True d True e False 9 Teacher to check; see figure 1 10 Exothermic – products at a lower energy level than reactants 11 Weigh the manganese dioxide at the beginning, carry out the reaction. Filter the mixture, dry and reweigh the manganese dioxide (filtrate). It should have the same mass as at the beginning. Could then also check to see if it catalyses the same reaction using fresh hydrogen peroxide 12 e.g. 0°C (273K) or 25°C (298K) and pressures around 101.3 kPa (1 atmosphere) 13 Lowering activation energy 14 e.g. 'Help' reactions take place, not used up in reaction 15 Enzymes more specific, easily broken down 16 Speed up reaction, use less energy, lower costs

27 Chemical equilibrium

1 a In an open system, solid calcium carbonate when heated changes to solid calcium oxide and carbon dioxide gas. b In a closed system, solid calcium oxide under pressure will react with carbon dioxide gas to form solid calcium carbonate. 2 No, some of the products will change back to reactants. 3 Because products are turning into reactants at the same rate as reactants are changing to products. 4 Micro: small; macro: large



5 6 Equilibrium 7 It doesn't matter whether you start with products (left to right) or reactants (right to left), the final equilibrium concentrations will be the same. 8 e.g. Change heat, pressure, add/remove chemicals 9 The forward reaction would speed up to produce more hydrogen iodide. 10 Equilibrium would be reached quicker because catalysts speed up both forward and reverse reactions by reducing activation energy. The position of equilibrium would be the same. 11 Add extra reactants, remove products 12 a Favour reactants/drive reaction to left b Favour products/drive reaction to right c Favour products/drive reaction to right d Favour reactants/drive reaction to left 13 a Favour products b Favour products c Favour products d No change (two volumes on both sides) 14 Teacher to check 15 A forecast of what may happen.

16 Laws are fundamental truths; theories are ideas that are used to explain things. Theories can be disproved.

28 Chemistry assignment

1 Teacher to check, but should include: a Discovery of neutron, England, ca 1935 b Atomic structure, Denmark, ca 1922 c Atomic theory, England, early 19th century d Discovered electron, atomic model, England, ca 1897 e Planetary model of the atom, England, ca 1911 f Periodic Table, Russia, ca 1871 g Avogadro's Law, Italy, ca 1811 h Various gas laws, France, early 19th century i Atomic theory, Greece, ca 400 bc j Spectroscopy, etc., Germany, middle of the 19th century 2 Teacher to check 3 a Electrically-charged particle b Part that makes up an atom, i.e. electron, proton, neutron c Particle made up of two or more atoms chemically combined d An atom of an element that has a different number of neutrons in the nucleus. e List of elements arranged in order of increasing atomic number f A large molecule made up of many repeating, similar molecules. g To do with carbon compounds h Donates protons, turns litmus red, pH less than 7 i Opposite of an acid, accepts protons, pH greater than 7 j The smallest part of an element that has the properties of that element. k An organic compound with –OH as the functional group l A logarithmic scale of hydrogen ion concentration. m Eight, e.g. the number of outer electrons in a noble gas n Avogadro's number, i.e. 6×10^{23} 4 a Six prizes are awarded annually for efforts in particular areas. b Alfred Nobel; in 1895 he left a will to start the awards. c To award those who had benefitted humanity d Teacher to check 5 Absolute zero is 0K or –273.16°C. A rise (or drop) of 1°C is the same as a rise (or drop) of 1K. The Kelvin was named after Lord Kelvin (William Thomson) who did much work in thermo-dynamics and many branches of physics. 6 When chemical reactions occur, there is no change in mass. Specific compounds always contain the same ratio of elements. 7 a iv b iii c ii d i 8 Teacher to check 9 Teacher to check; should include that chemists study the behaviour of chemicals and pharmacists dispense medicines

29 Chemistry crosswords

Metals Across 3 Calcium 5 Iron 6 Magnesium 8 Silver 9 Aluminium 12 Copper 13 Sodium 14 Gold **Down** 1 Tin 2 Potassium 4 Cadmium 7 Vanadium 10 Nickel 11 Lead

Non-metals Across 6 Oxygen 7 Sulfur 9 Phosphorus 10 Nitrogen 12 Bromine 13 Iodine **Down** 1 Carbon 2 Argon 3 Fluorine 4 Hydrogen 5 Helium 8 Chlorine 11 Neon

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the **highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Assignments in Chemistry Book 2

This new three-book chemistry series by highly-experienced science authors and teachers Peter Gribben and Wendy St Germain provides photocopiable worksheets designed to challenge and stimulate students, as they explore a variety of chemistry concepts.

There are an array of opportunities for students to test their scientific skills through the analysis and interpretation of diagrams, tables and graphs. Students will also extend their reasoning and literacy skills as they are guided through the application of theoretical concepts to practical examples.

Research tasks are included to expand students' knowledge of the history of chemistry and the contributions of individual scientists.

Each assignment is self-contained and requires minimal teacher preparation. Answers have been provided for ease of marking.

Topics include

Moles and gases
Concentrations of solutions
Solutions, suspensions, colloids and emulsions
Solubility and precipitation
Electronegativity
Periodic properties
Dependability of information

The extraction, refining, properties and uses of metals
Water quality and oxygen
Viscosity
Saponification
Energy and reactions
Chemical equilibrium

Associated titles in the Instant Lessons series

- Assignments in Chemistry (Books 1 & 3)
- Instant Lessons in Chemistry (Books 1–3)
- Interpreting Data in Chemistry (Books 1–2)
- Assignments in Genetics (Books 1–2)
- Assignments in Biology (Books 1–3)
- Assignments in Physics (Books 1–2)